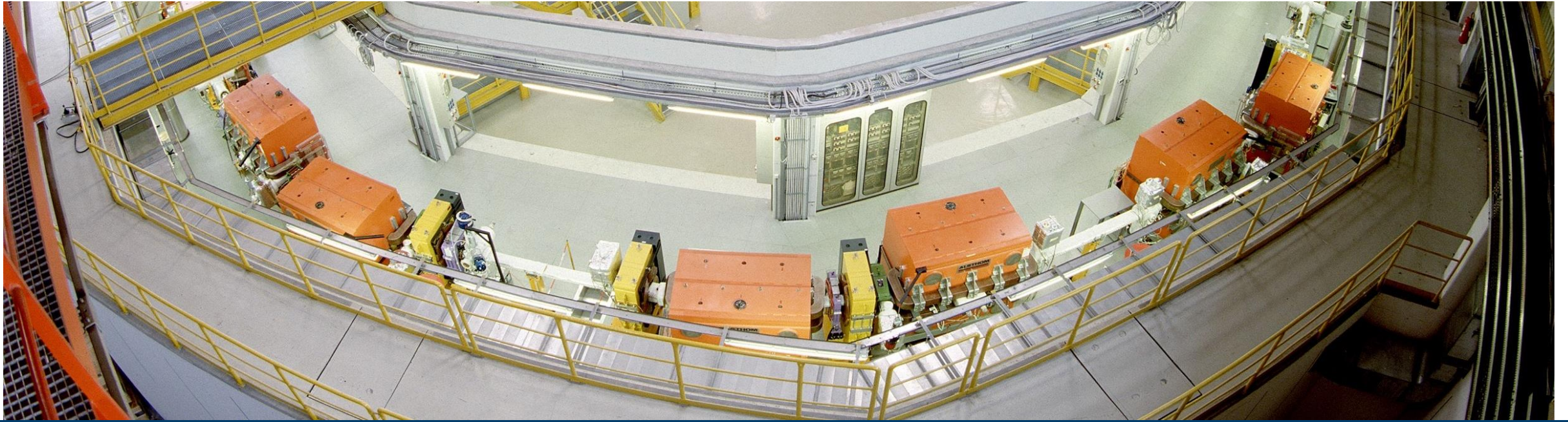




THE SEARCH FOR ELECTRIC DIPOLE MOMENTS OF CHARGED PARTICLES IN STORAGE RINGS



DPG Erlangen

19.03.2026 | ACHIM ANDRES



MATTER ANTIMATTER ASYMMETRY

- Big Bang: Equal amount of matter and antimatter

$$N_B = N_{\bar{B}}$$

- Early Universe: Asymmetric annihilation process

$$\text{Theory: } B + \bar{B} \rightarrow \gamma + \gamma + \dots$$

$$\text{Measurement: } B + \bar{B} \rightarrow \gamma + \gamma + \dots + \textcolor{red}{B} + \dots?$$

- Today: Asymmetry between matter and antimatter

$$\eta = \frac{N_B - N_{\bar{B}}}{N_\gamma}$$

- Mismatch between expectation and measurement

$$\eta^{\text{SCM}} \approx 10^{-18} \quad \text{versus} \quad \eta^{\text{meas}} \approx 10^{-10}$$



- Andrei Sakharov (1976)
 - Baryon Number Violation
 - **C and CP Violation**
 - Deviation from thermal equilibrium

PERMANENT EDMS AND CP VIOLATION

- EDM is a **vectorial** property aligned with **the particles' spin**
- Magnetic Dipole Moment (MDM): $\vec{\mu} = \mu \frac{\vec{s}}{|\vec{s}|}$ with $\mu = g \frac{q\hbar}{2m}$
- Electric Dipole Moment (EDM): $\vec{d} = d \frac{\vec{s}}{|\vec{s}|}$ with $d = \eta^{\text{EDM}} \frac{q\hbar}{2mc}$

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$$H = -\vec{d} \cdot \vec{E} - \vec{\mu} \cdot \vec{B}$$

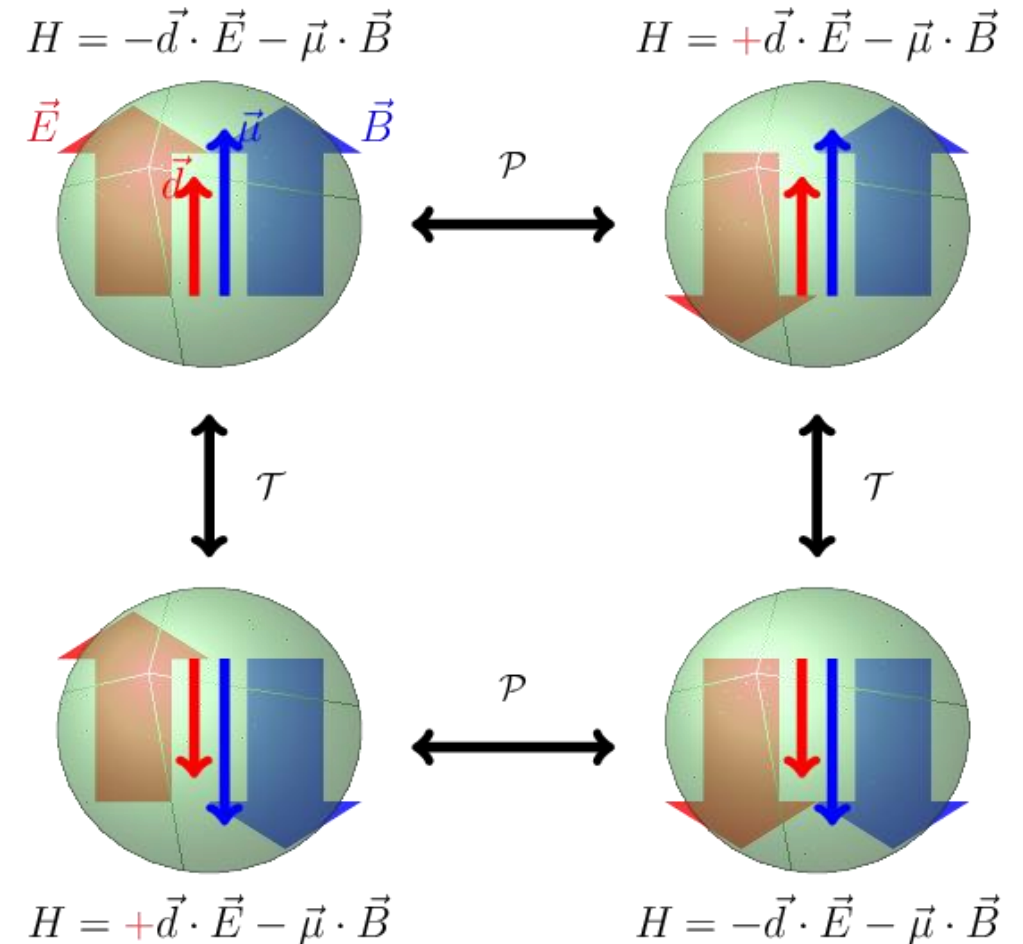
$$\text{Parity } P: H = +\vec{d} \cdot \vec{E} - \vec{\mu} \cdot \vec{B}$$

$$\text{Time } T: H = +\vec{d} \cdot \vec{E} - \vec{\mu} \cdot \vec{B}$$

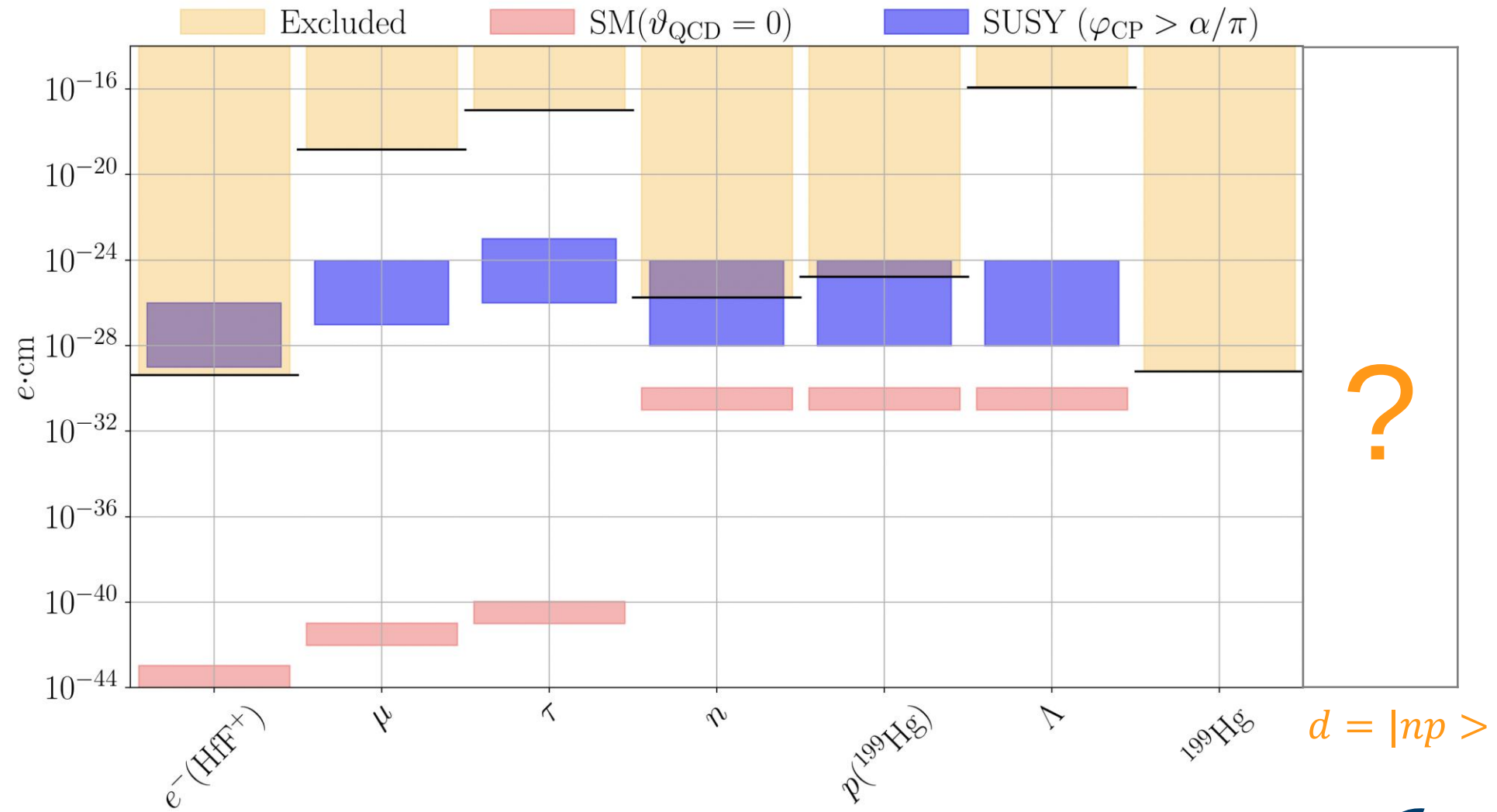
- According to CPT - Theorem:

$$T \text{ Violation} = CP \text{ Violation}$$

⇒ EDM violates both *P* and *CP* symmetry



PERMANENT EDM SEARCH



ELECTRIC DIPOLE MOMENTS

Axion Search

- Violation of symmetries was observed in the weak sector
- **CP** violation in the strong sector

$$L_{\bar{\theta}_{\text{QCD}}} = -\bar{\theta}_{\text{QCD}} \frac{g_s^2}{64\pi^2} \varepsilon^{\mu\nu\alpha\beta} G_{\mu\nu}^a G_{\alpha\beta}^a$$

- Limits from neutron EDM measurements limit constrain

$$\bar{\theta}_{\text{QCD}} \leq 10^{-10} \Rightarrow \text{Strong CP Problem}$$

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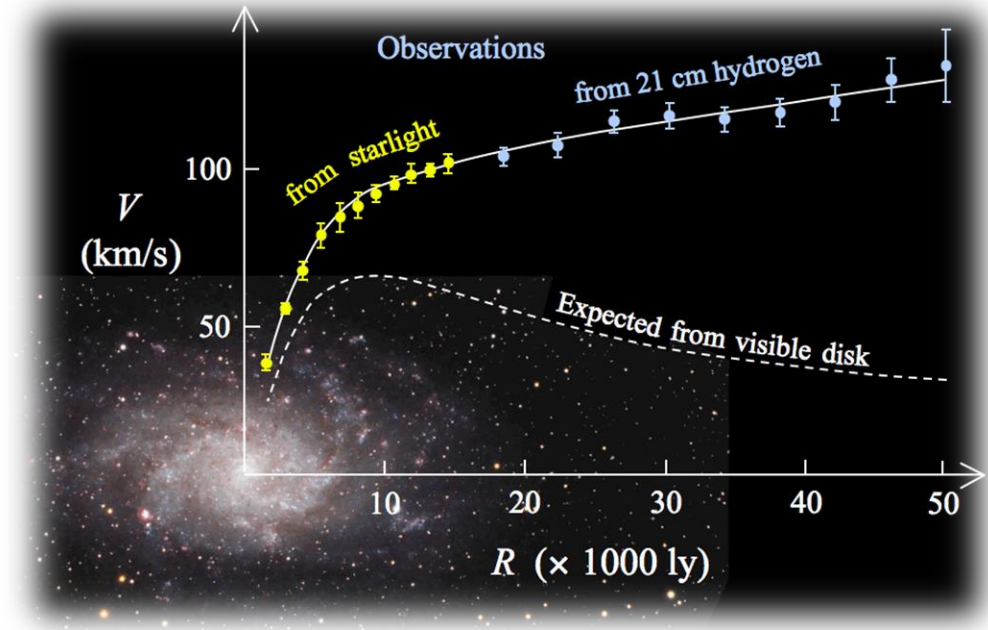
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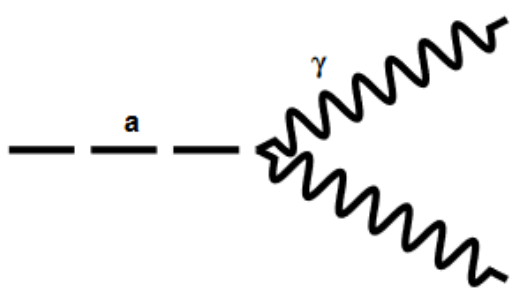
- Problem solver: **Axion**

$$\vec{d} = d \frac{\vec{s}}{|\vec{s}|} \quad \text{with } d = d_{\text{P}} + d_{\text{Osc}} \cos(\omega_a + \phi_a) \quad \text{and } \omega_a = \frac{m_a c^2}{\hbar}$$

- Existence of an axion leads to an additional oscillating EDM component
- Axion could explain the strong CP - problem
- Axions are potential candidates for Dark Matter



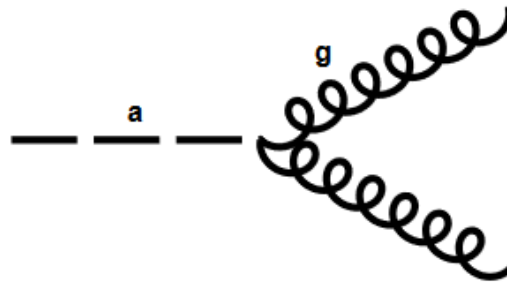
AXION COUPLING

$$\mathcal{L} : -\frac{\alpha}{8\pi} \frac{C_\gamma}{f_a} \mathbf{a} F_{\mu\nu} \tilde{F}^{\mu\nu}$$


A Feynman diagram showing a horizontal dashed line labeled 'a' (axion) on the left, which then splits into two wavy lines labeled 'γ' (photons) extending to the right.

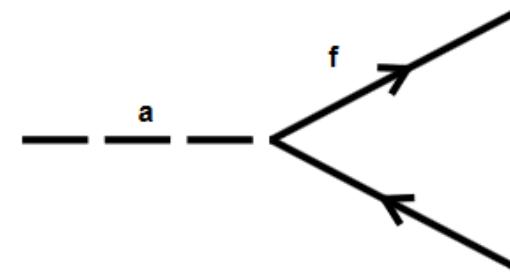
studied by many experiments

$$-\frac{\alpha_s}{8\pi} \frac{C_G}{f_a} \mathbf{a} G_{\mu\nu}^b \tilde{G}^{b,\mu\nu}$$



Electric Dipole Moment (EDM)

$$-\frac{1}{2} \frac{C_N}{f_a} \partial_\mu \mathbf{a} \bar{\psi}_f \gamma^\mu \gamma^5 \psi_f$$



axion wind term

**accessible in storage ring experiments
with spin polarized beams**

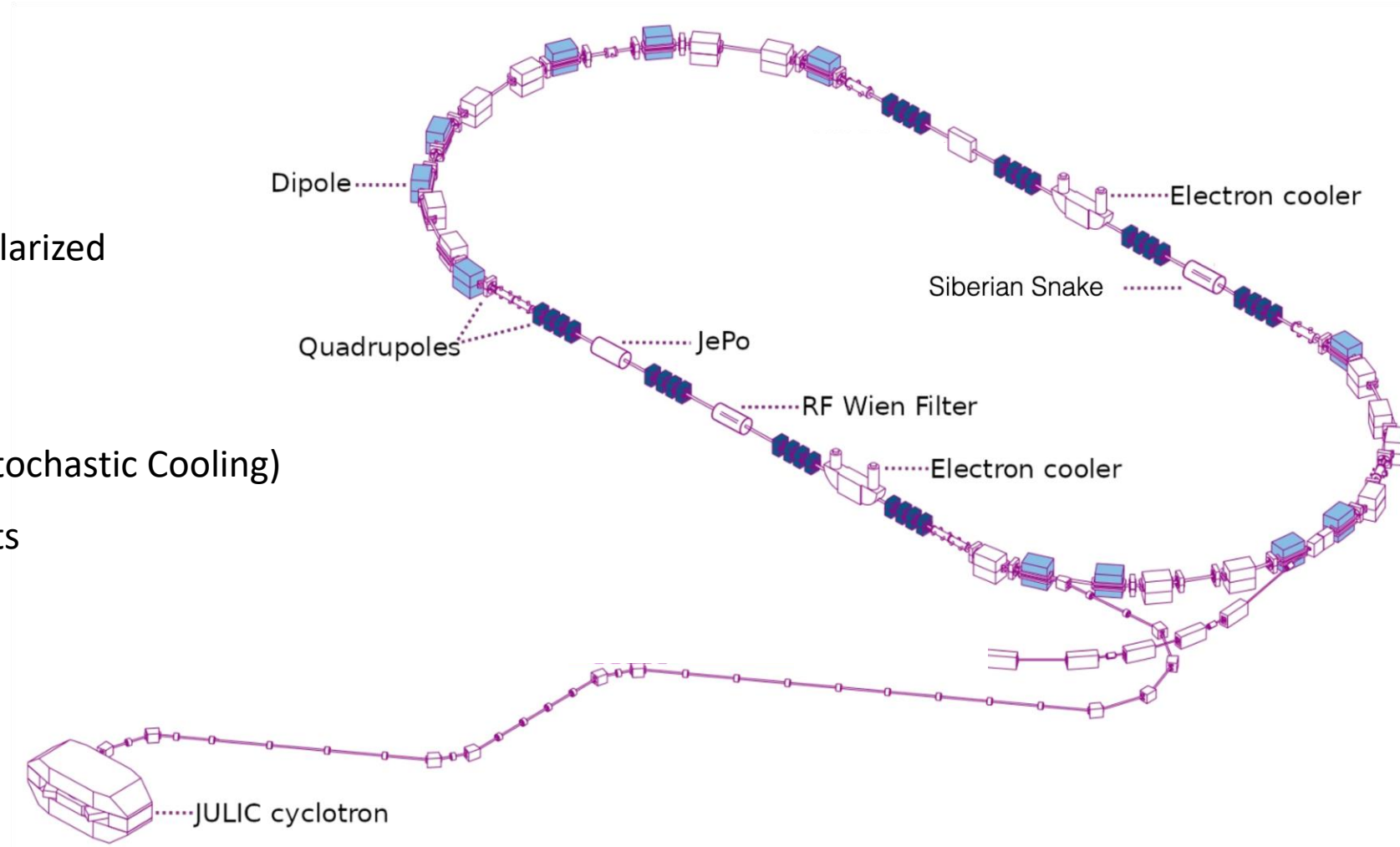
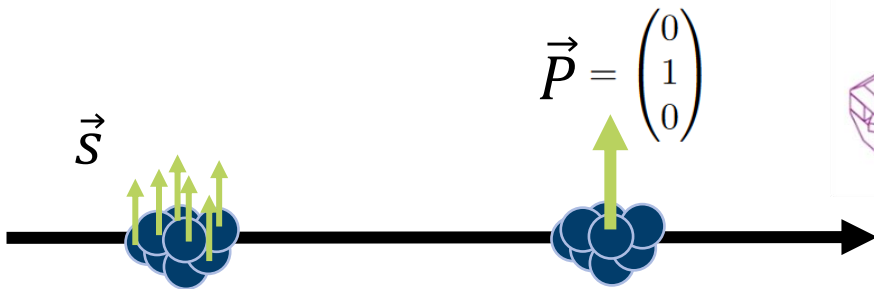
COSY - COOLER SYNCHROTRON (1993 – 2023)

Overview

- Magnetic storage ring
- Circumference 184 m
- Accelerate and Store **Polarized** / Unpolarized

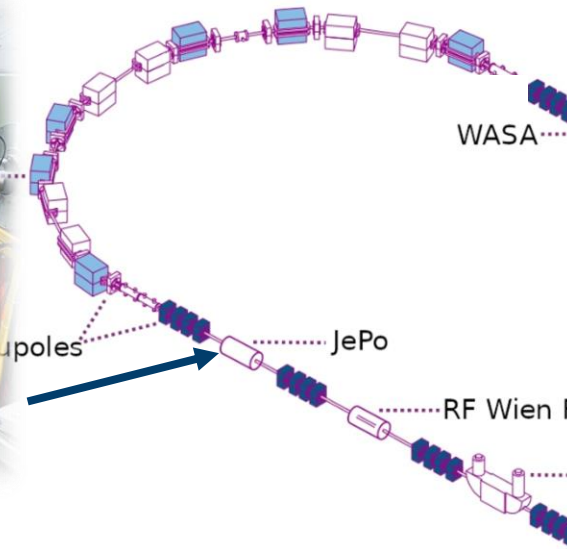
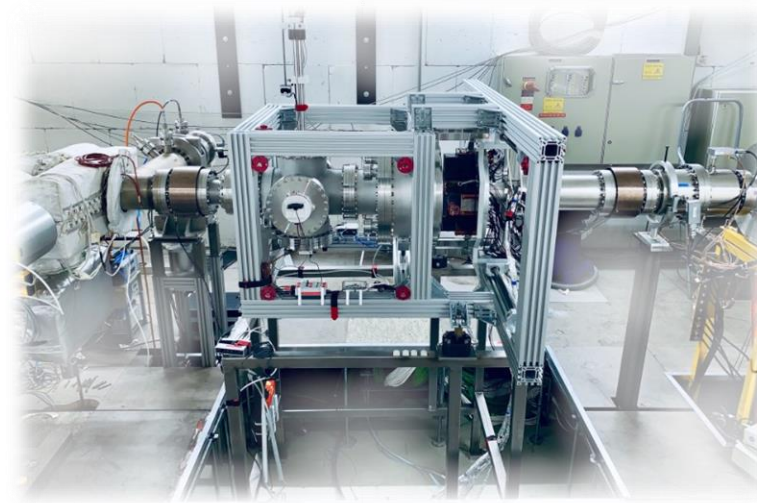
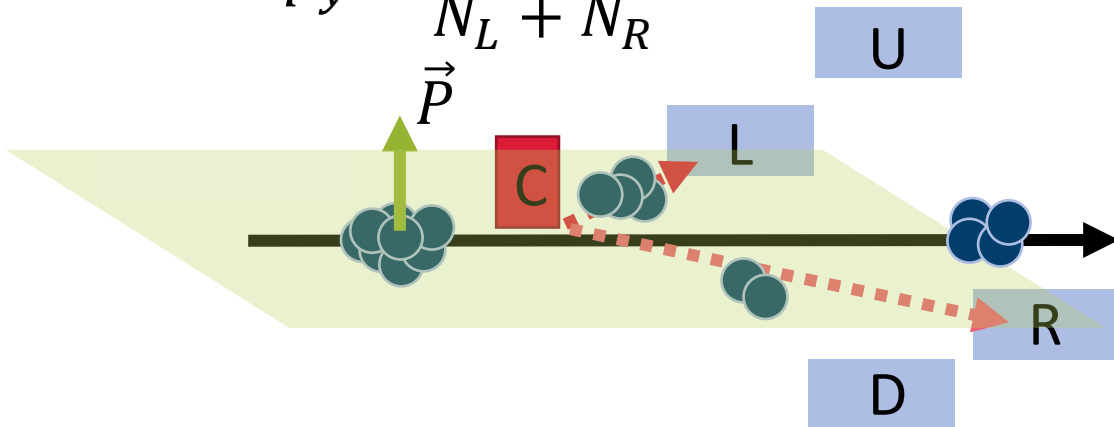
Deuterons and Protons

- $p = 0.3 - 3.7 \text{ GeV}/c$
- Excellent Beam Quality (e-Cooling & Stochastic Cooling)
- Hadron Physics / **Precision** Experiments



POLARIMETER

$$p_y \propto \frac{N_L - N_R}{N_L + N_R}$$



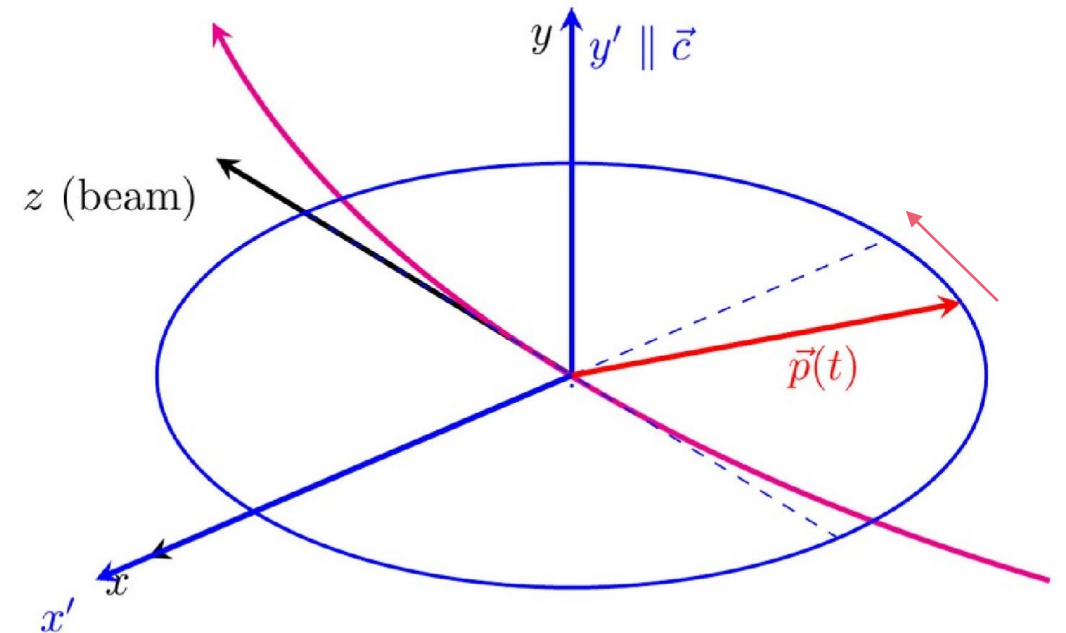
SPIN DYNAMICS

- Thomas-BMT Equation

$$\frac{d\vec{S}}{dt} = (\vec{\Omega}_{\text{MDM}} + \vec{\Omega}_{\text{EDM}}) \times \vec{S}$$

$$\vec{\Omega}_{\text{MDM}} = -2\pi\gamma G f_{\text{rev}} \vec{e}_y \quad (G_d = -0.143)$$

- All Experiments start with in-plane polarization



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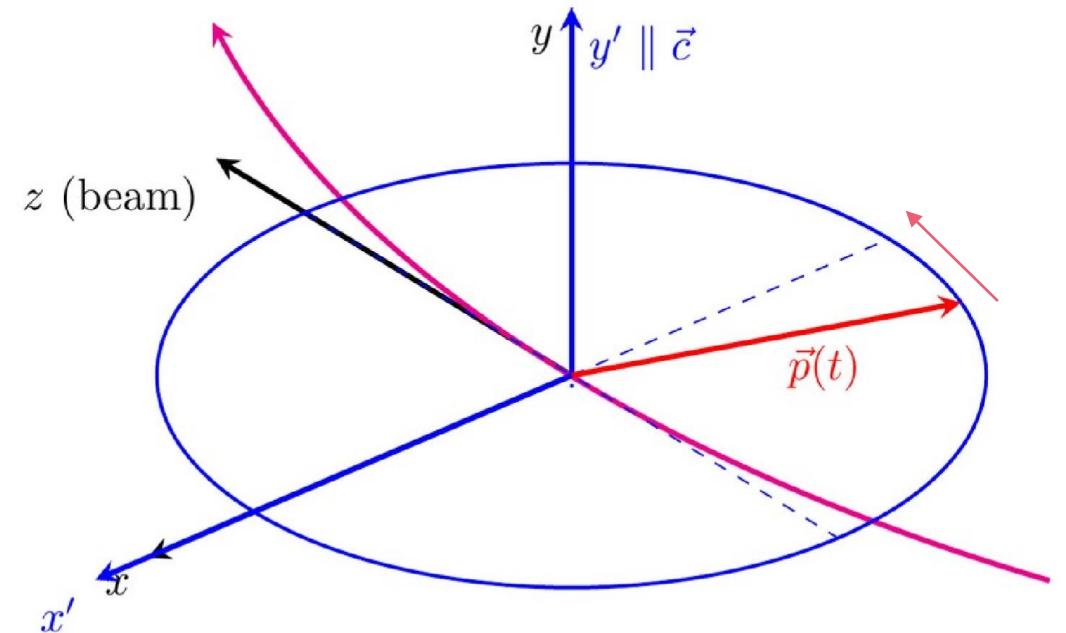
- All Experiments start with in-plane polarization

- $f_s = \gamma G f_{\text{rev}} \approx 121 \text{ kHz}$ (Deuterons @ 970 MeV)

- $\sigma_{f_s} \approx 10^{-10}$ in 100 s

- $\sigma_{f_s} \approx 10^{-8}$ in 2 s

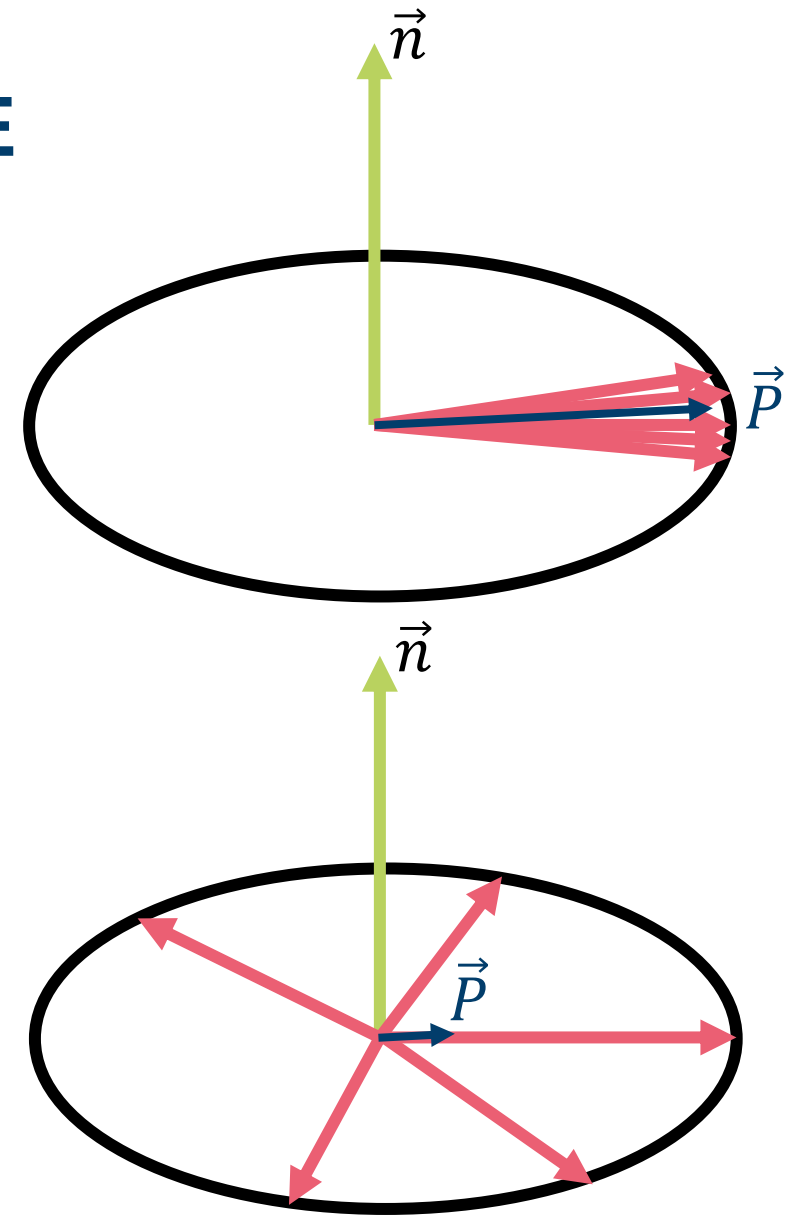
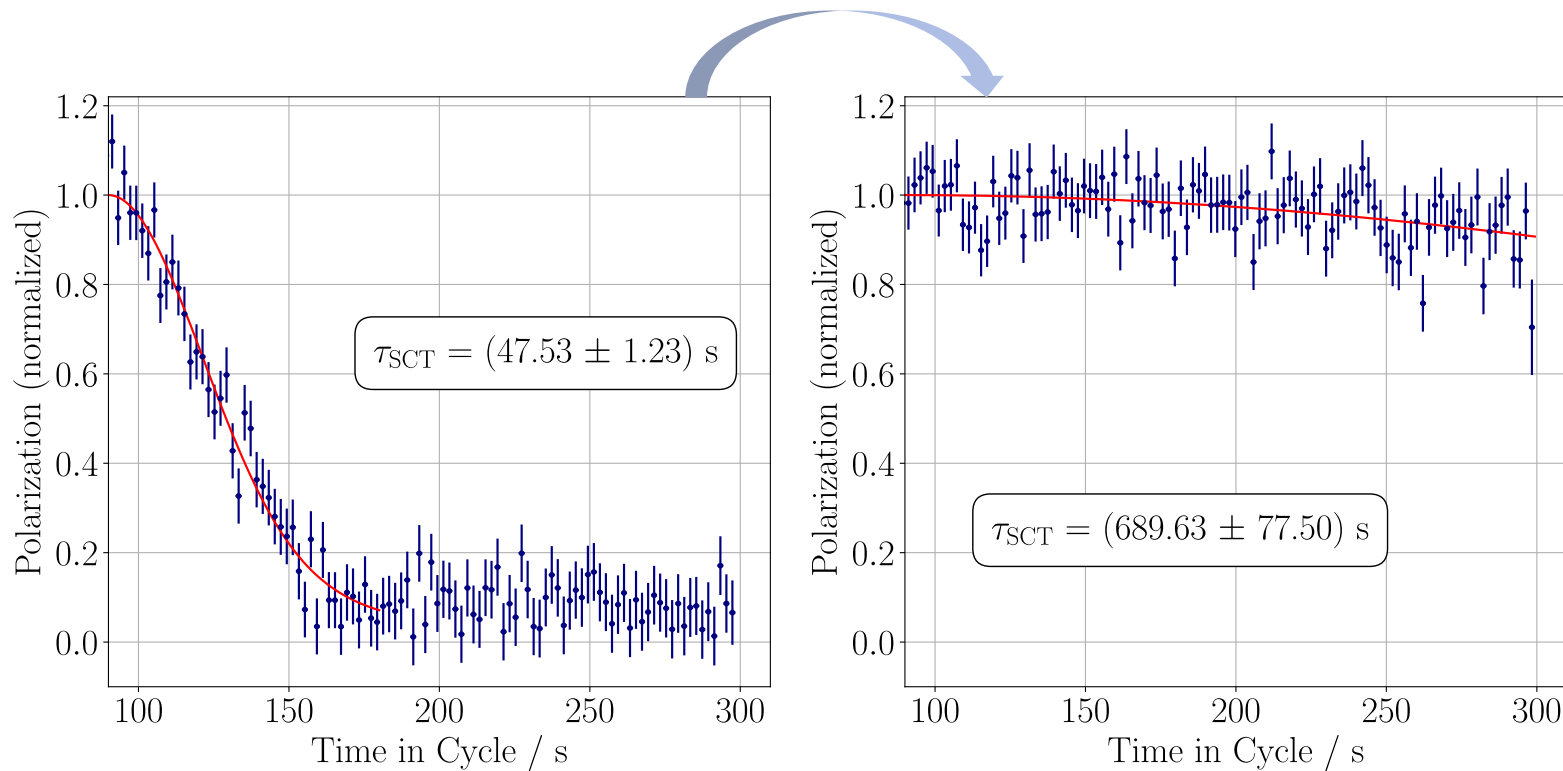
[PRL 115, 094801 \(2015\)](#)



PREREQUISITE: SPIN COHERENCE TIME

- Spin oscillation frequency: $\vec{\Omega}_{\text{MDM}} = -2\pi\gamma G f_{\text{rev}} \vec{e}_y$

sextupole optimization



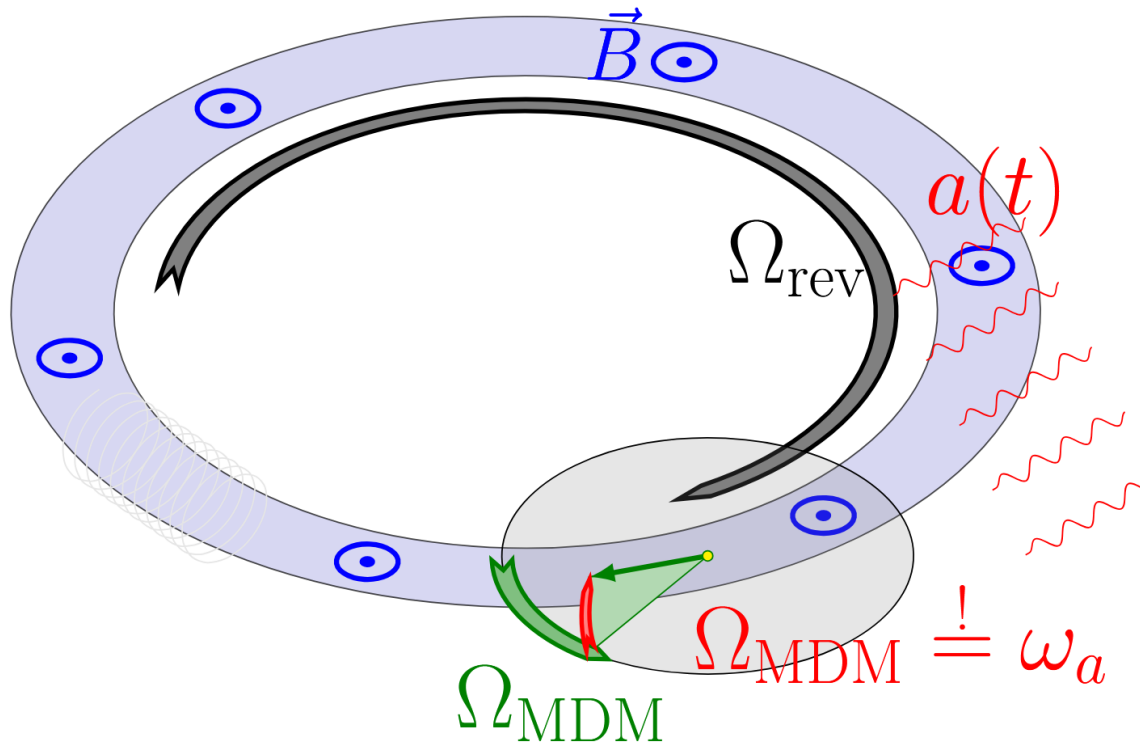
$$d = d_P + d_{\text{Osc}} \cos(\omega_a + \phi_a)$$

PRINCIPLE OF STORAGE RING AXION EXPERIMENTS

$$d = d_p + d_{osc} \cos(\omega_a + \phi_a)$$

$$\vec{\Omega}_{\text{EDM}} = -\frac{1}{S\hbar} d(t) c \vec{\beta} \times \vec{B},$$

$$\vec{\Omega}_{\text{wind}} = -\frac{1}{S\hbar} \frac{C_N}{2f_a} [\hbar \partial_0 a(t)] \vec{\beta},$$

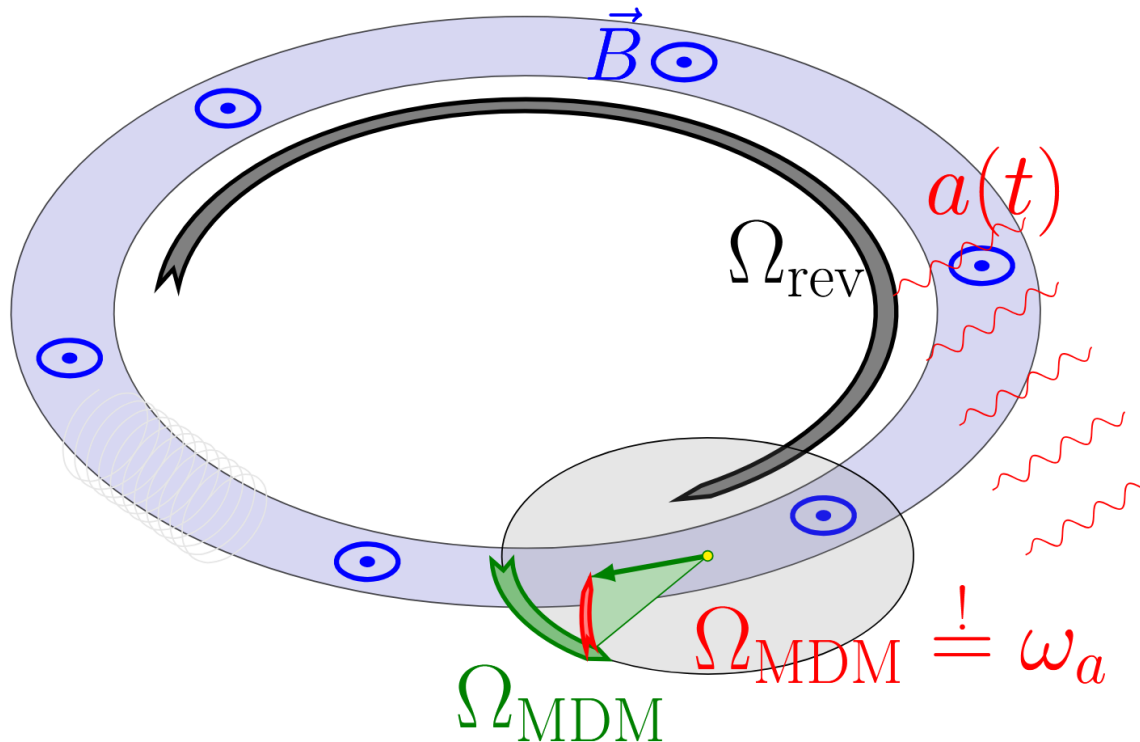


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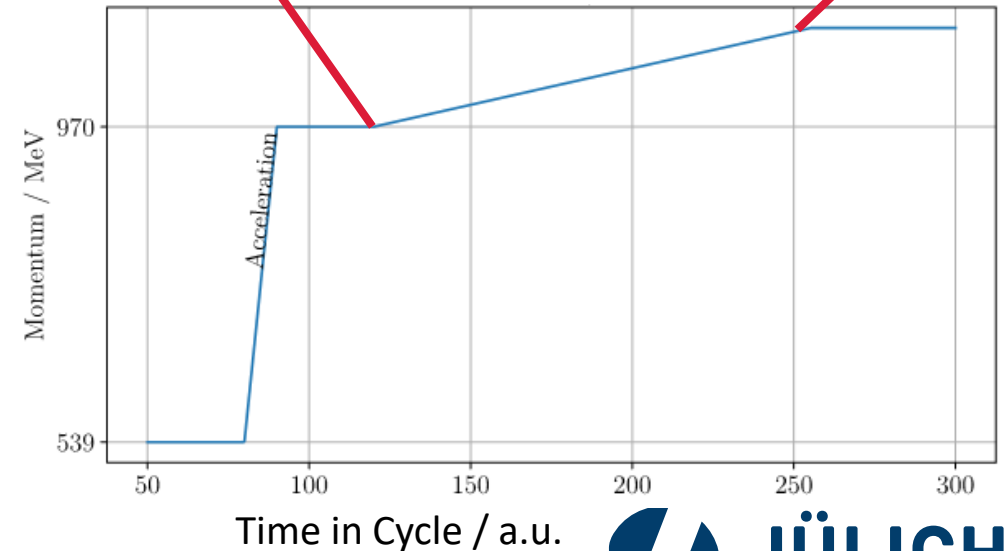
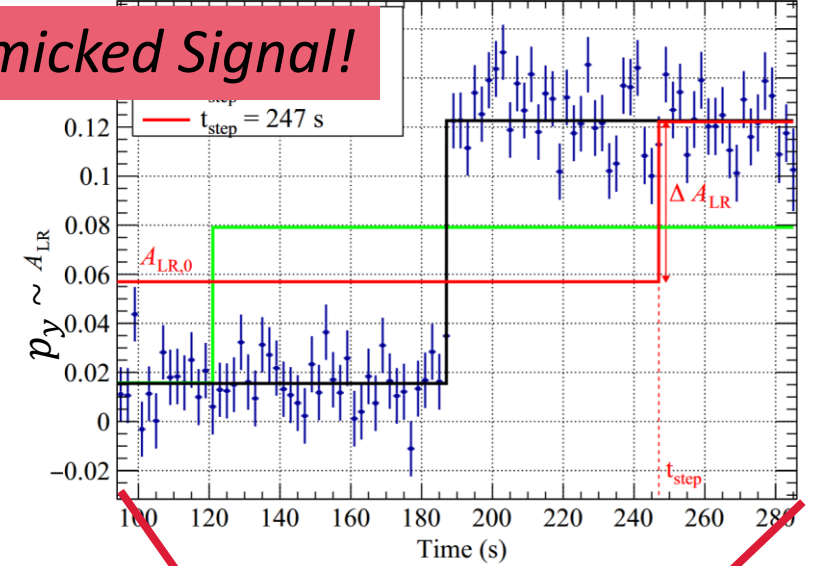
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Mimicked Signal!

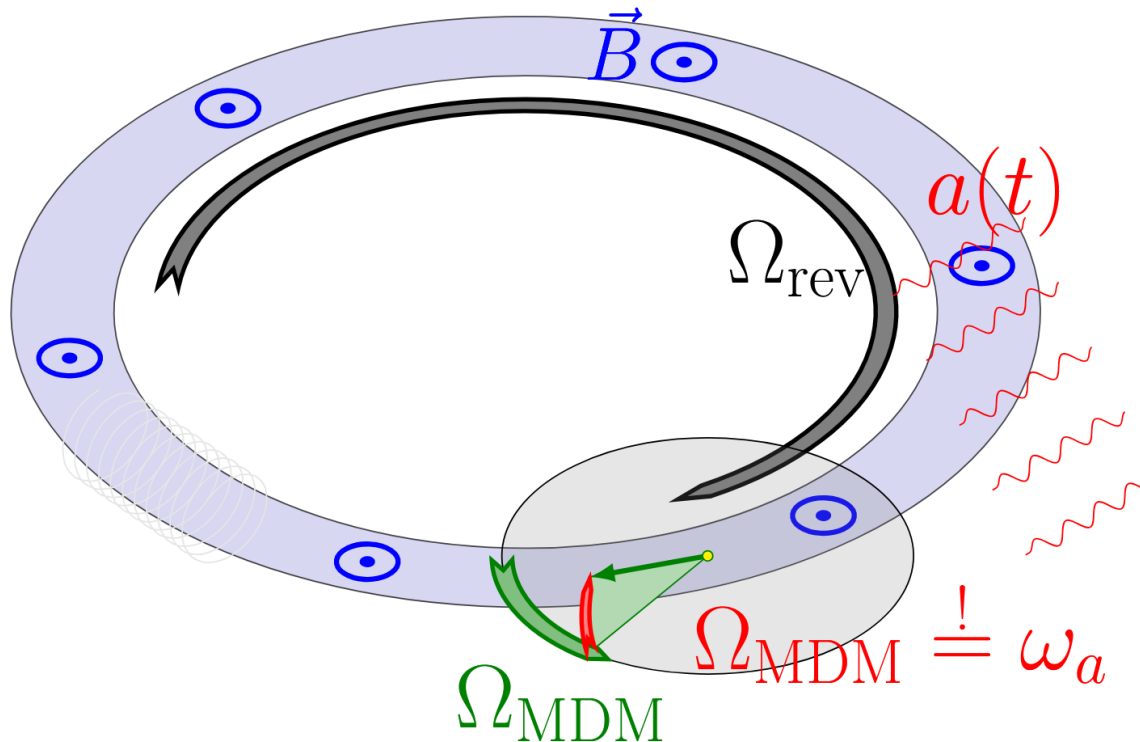


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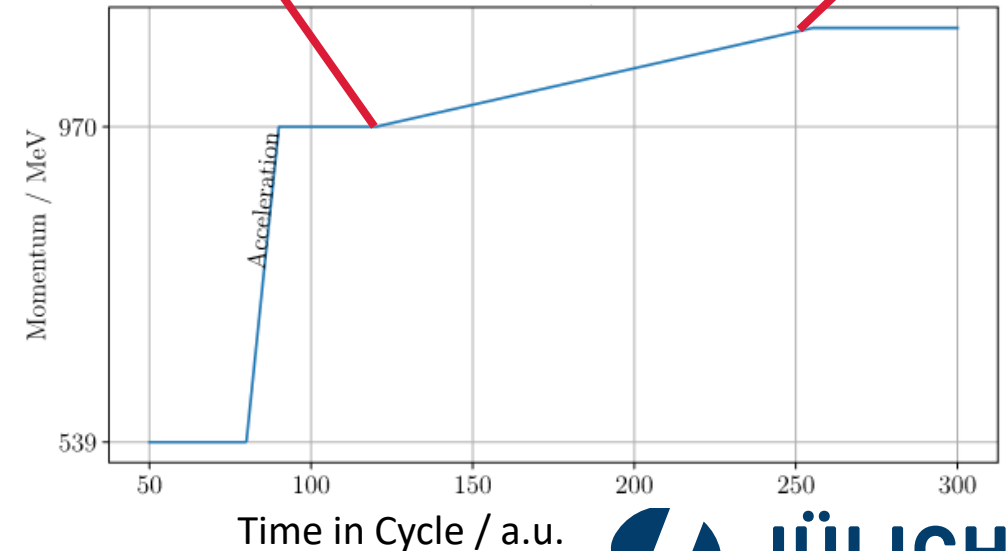
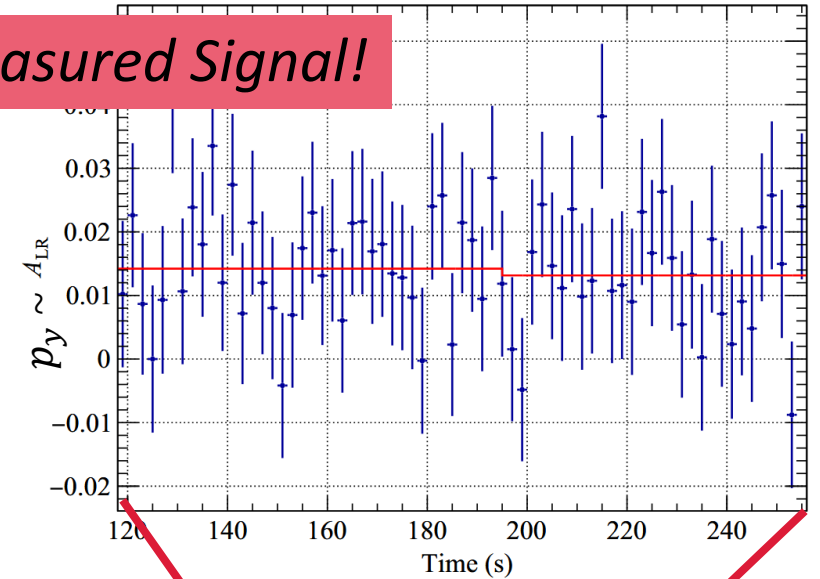
$$d = d_p + d_{osc} \cos(\omega_a + \phi_a)$$

Scanned Frequency Range: 120.05 – 121.45 kHz

Covered Axion Mass Range: $(4.96 - 5.02) \cdot 10^{-10} \text{ eV}/c^2$



Measured Signal!

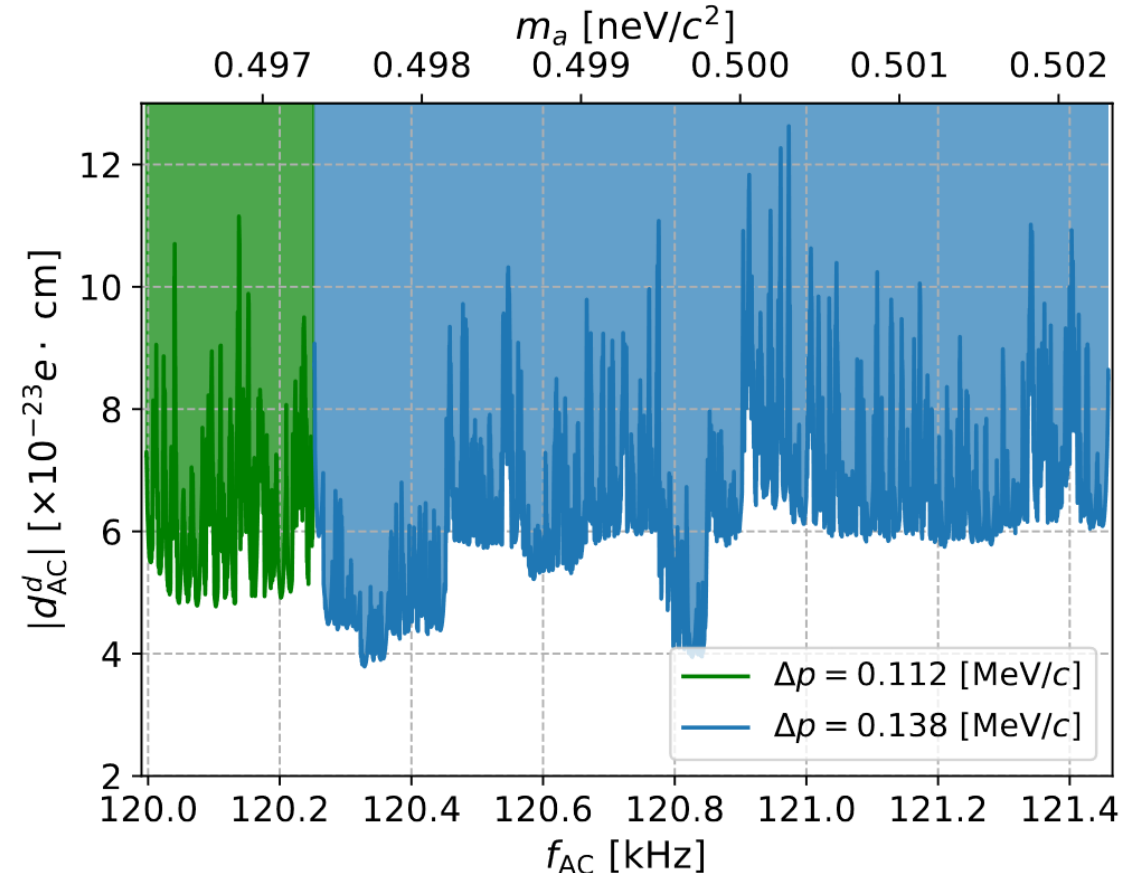


CONCLUSION: AXION SEARCH

$$d = d_p + d_{\text{OSC}} \cos(\omega_a + \phi_a)$$

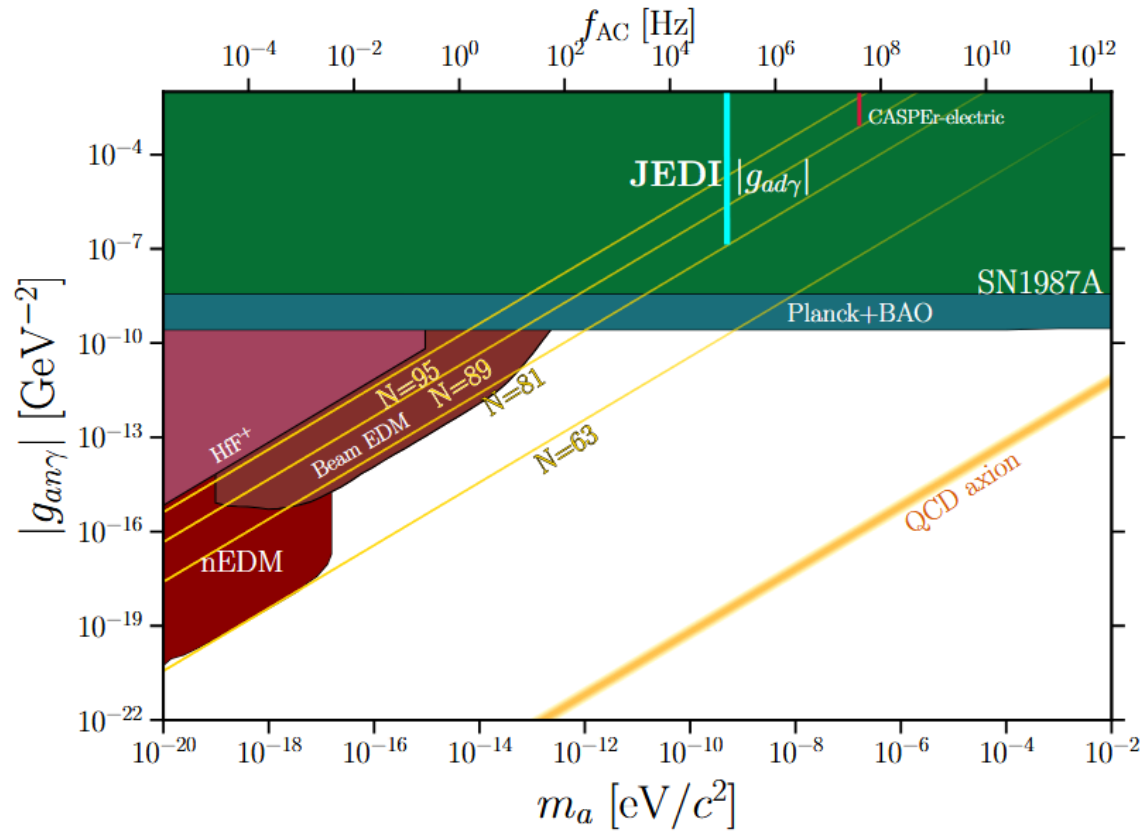
$$\omega_a = \frac{m_a c^2}{\hbar}$$

- $f_{\text{AC}} = \frac{1}{2\pi} \frac{m_a c^2}{\hbar} = \gamma G f_{\text{rev}}$
- Upper limit on the ALPs induced oscillating EDM
- Average of $|d_{\text{OSC}}| < 6.4 \times 10^{-23} e \cdot \text{cm}$
- Engineering Run of a few days
 - Increase scan range
 - Sit on the same frequency for longer time

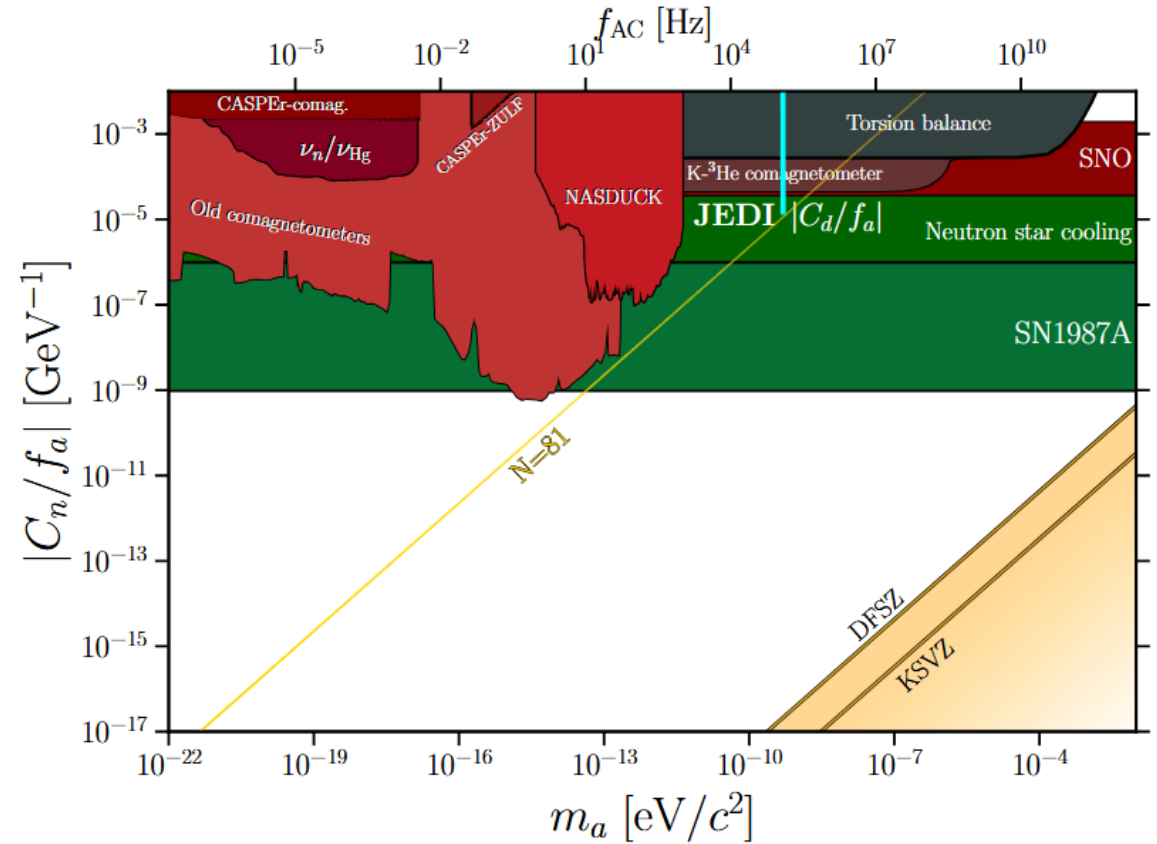


AXION COUPLING CONSTANTS

$$d = d_p + d_{osc} \cos(\omega_a + \phi_a)$$



Coupling to Gluons



Coupling to Nucleons

$$d = d_{\text{P}} + d_{\text{Osc}} \cos(\omega_a + \phi_a)$$

SPIN DYNAMICS

$$d = d_p + d_{osc} \cos(\omega_a + \phi_a)$$

$$\vec{d} = d \frac{\vec{s}}{|\vec{s}|} \text{ with } d = \eta^{\text{EDM}} \frac{q\hbar}{2mc}$$

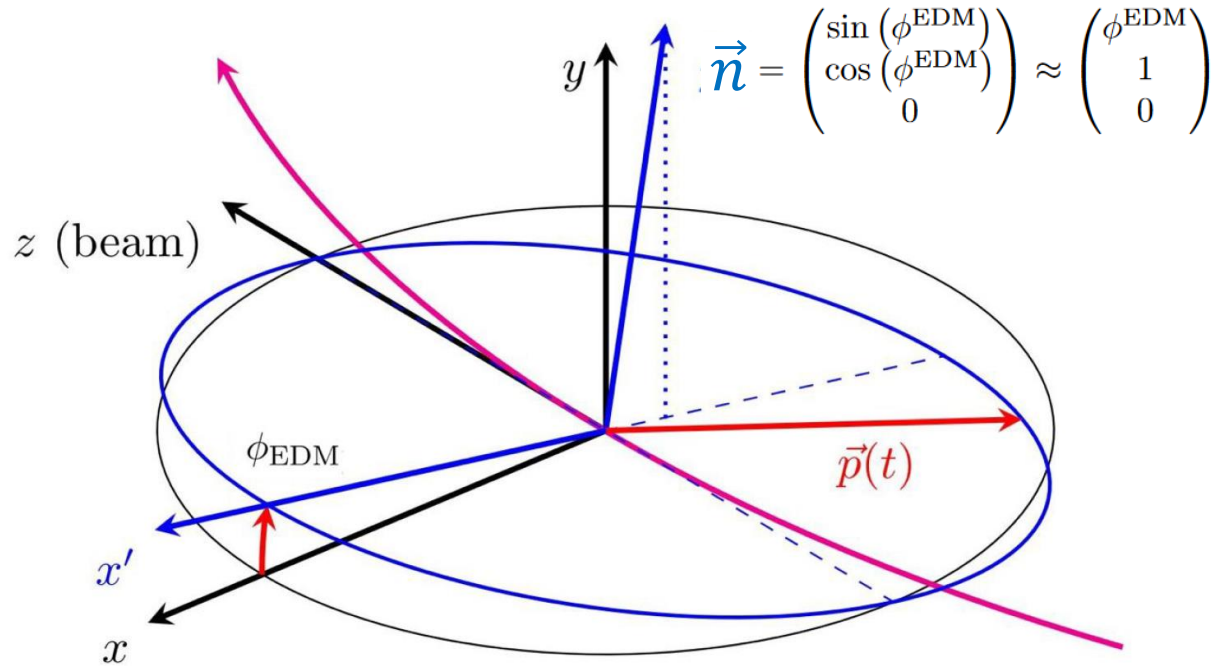
- Measure the influence of the EDM on the spin motion

$$\frac{d\vec{s}}{dt} = (\vec{\Omega}_{\text{MDM}} + \vec{\Omega}_{\text{EDM}}) \times \vec{s}$$

$$\vec{\Omega}_{\text{MDM}} = -2\pi\gamma G f_{\text{rev}} \vec{e}_y$$

$$\vec{\Omega}_{\text{EDM}} = \eta^{\text{EDM}} \gamma \beta f_{\text{rev}} \vec{e}_x$$

$$\arctan\left(\frac{|\vec{\Omega}_{\text{EDM}}|}{|\vec{\Omega}_{\text{MDM}}|}\right) \approx \phi_{\text{EDM}} \approx \eta^{\text{EDM}} \frac{2\beta}{G}$$



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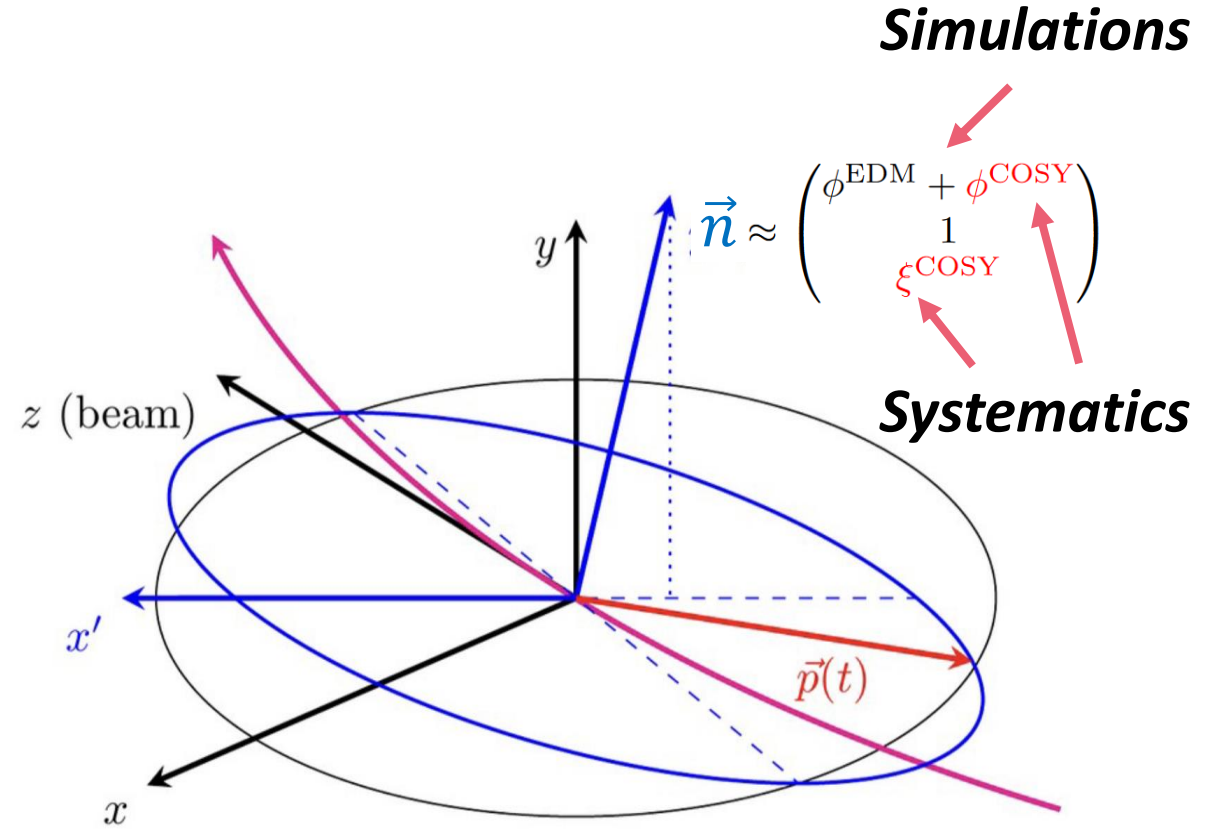
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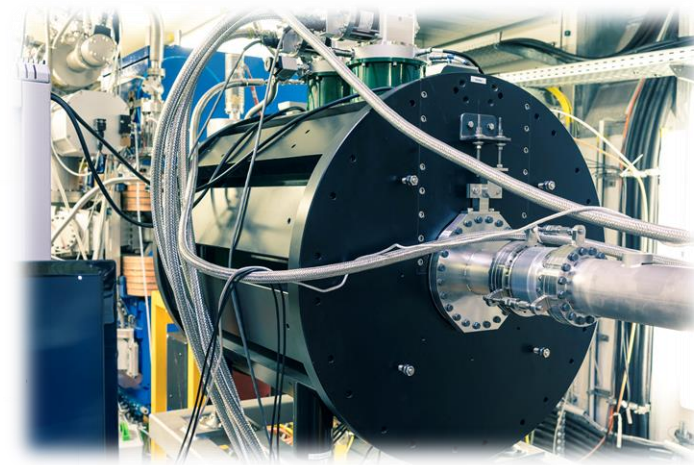
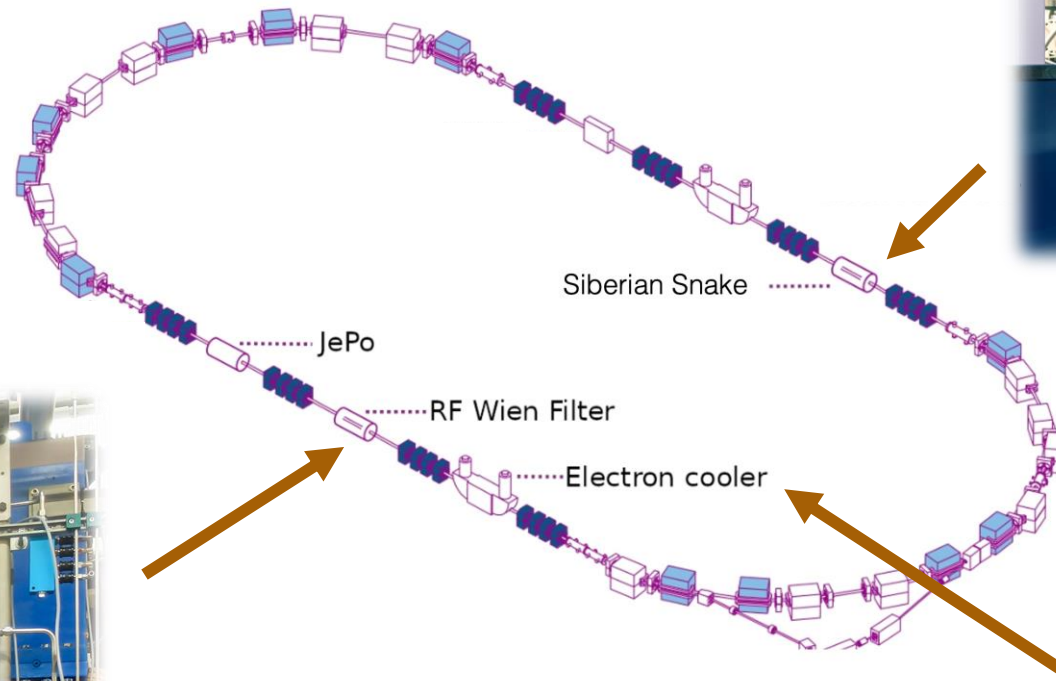
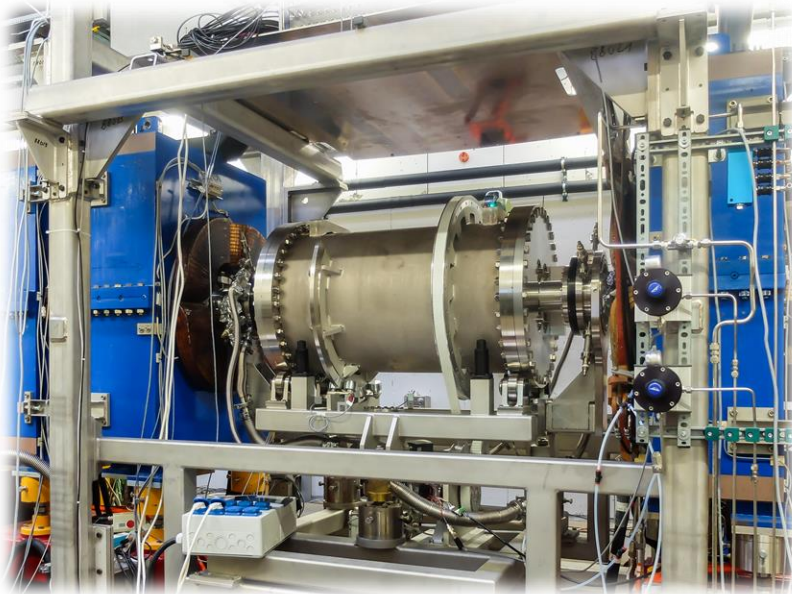
- **Problem:** Ring **imperfections** (magnet misalignments,..)

lead to rotations of $\hat{n}$ in **radial** (x) and **longitudinal** (z) direction



COSY

RF Wien Filter



$\hat{n}_{z, \text{Snake}}$

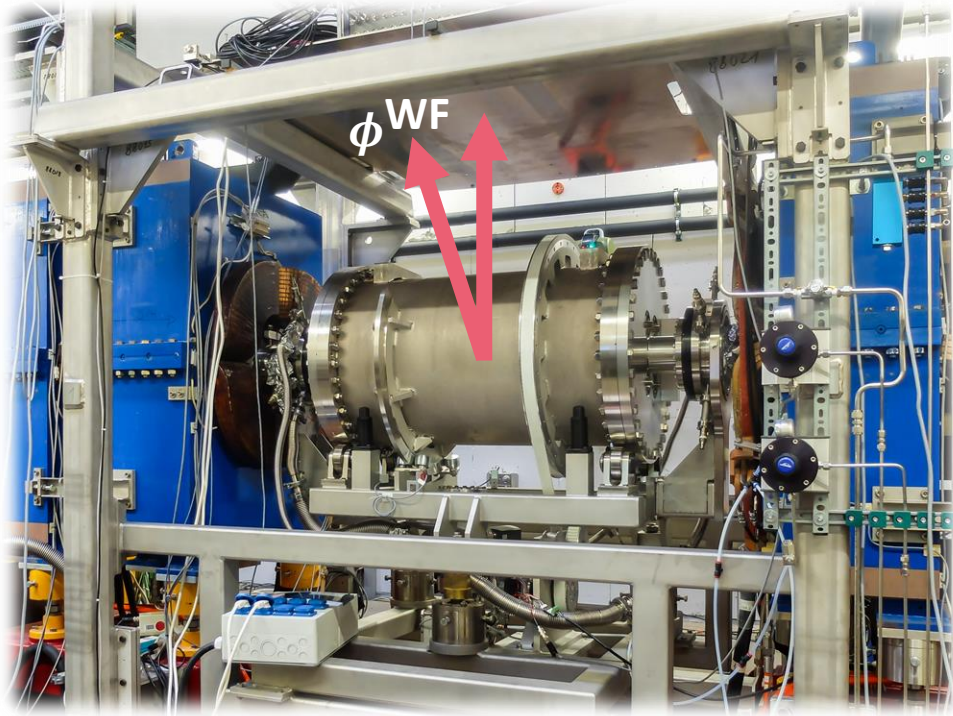
Independent!

$\hat{n}_{z, \text{Sol}}$



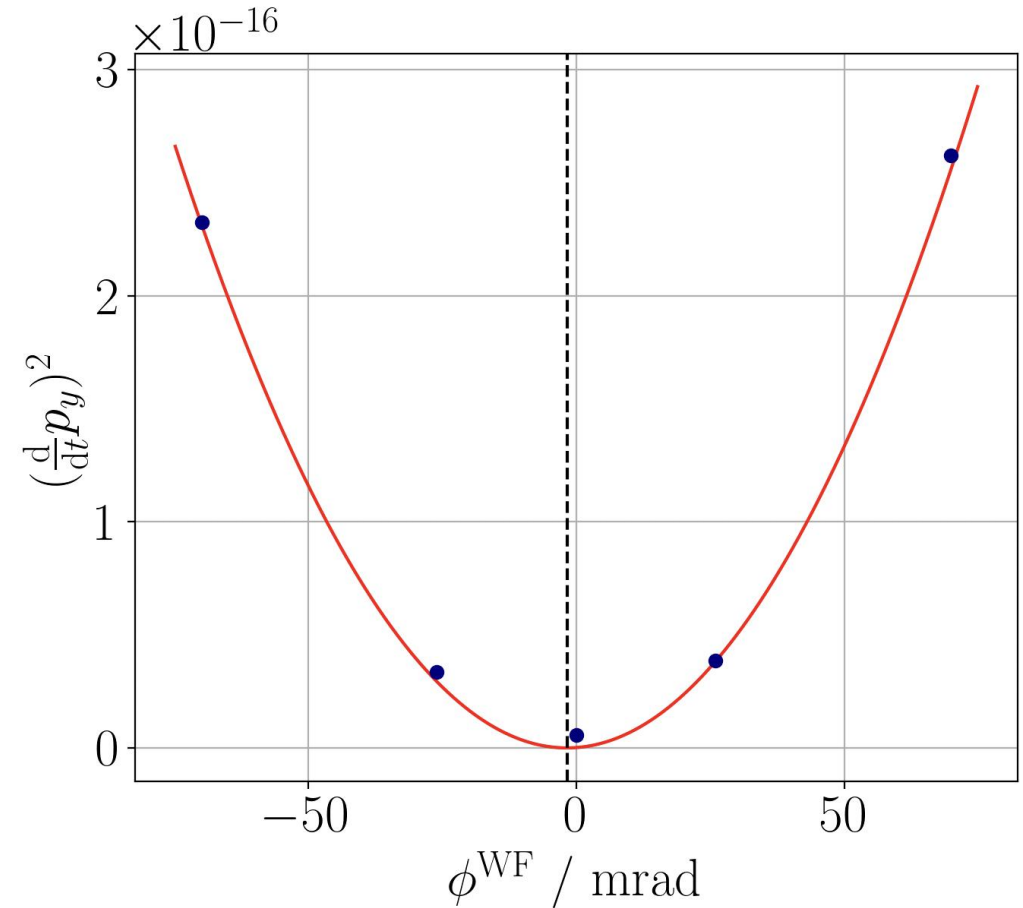
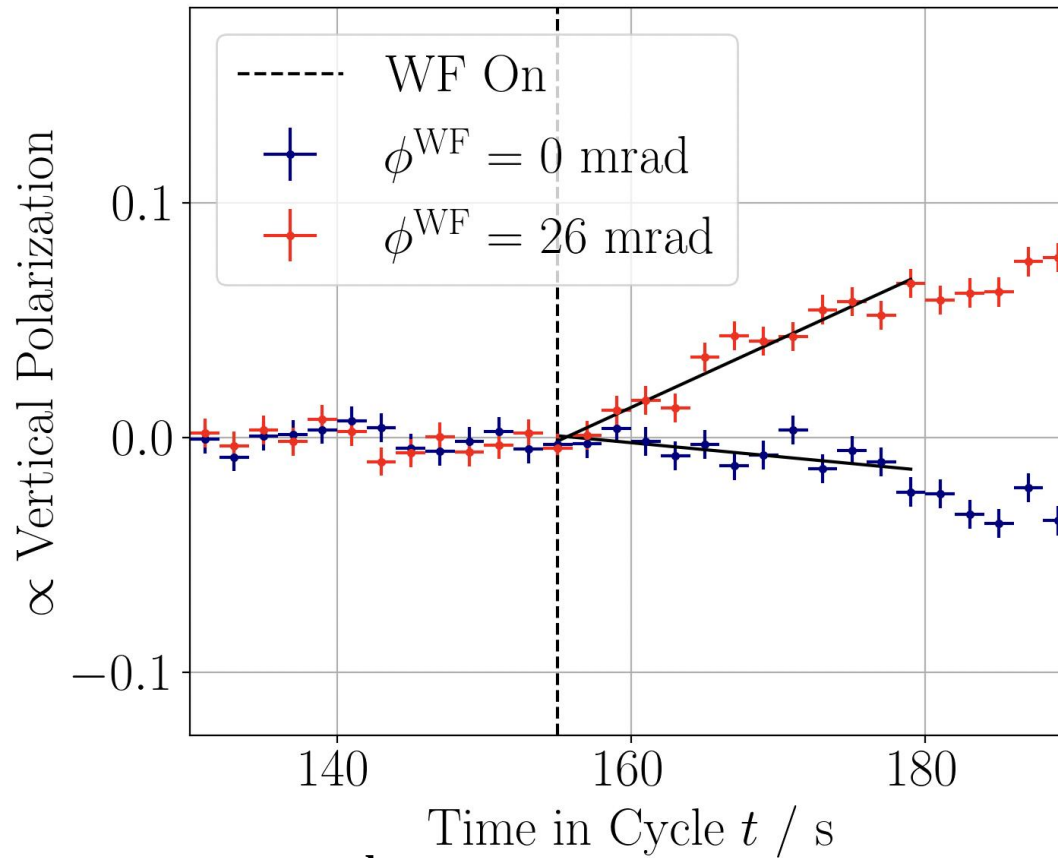
$\hat{n}_{x, \text{WF}}$ & $\hat{n}_{z, \text{WF}}$

RF WIEN FILTER



- Goal: **Measure n_x**
- $\vec{E} \perp \vec{B} \perp \text{Beam} \rightarrow \vec{F}_L = q \cdot (\vec{E} + \vec{v} \times \vec{B}) = 0$
- $\vec{B}$ - Field can be rotated around the beam pipe by ϕ^{WF}
- Needs to run on the spin precession frequency
$$f_{\text{WF}} = f_s$$
- $\vec{E} = \vec{E}_0 \cos(2\pi f_s + \phi)$ and $\vec{B} = \vec{B}_0 \cos(2\pi f_s + \phi)$
- Both frequencies need to have an adjustable phase relation
 - Phase feedback [PRAB 28, 062801 \(2025\)](#)

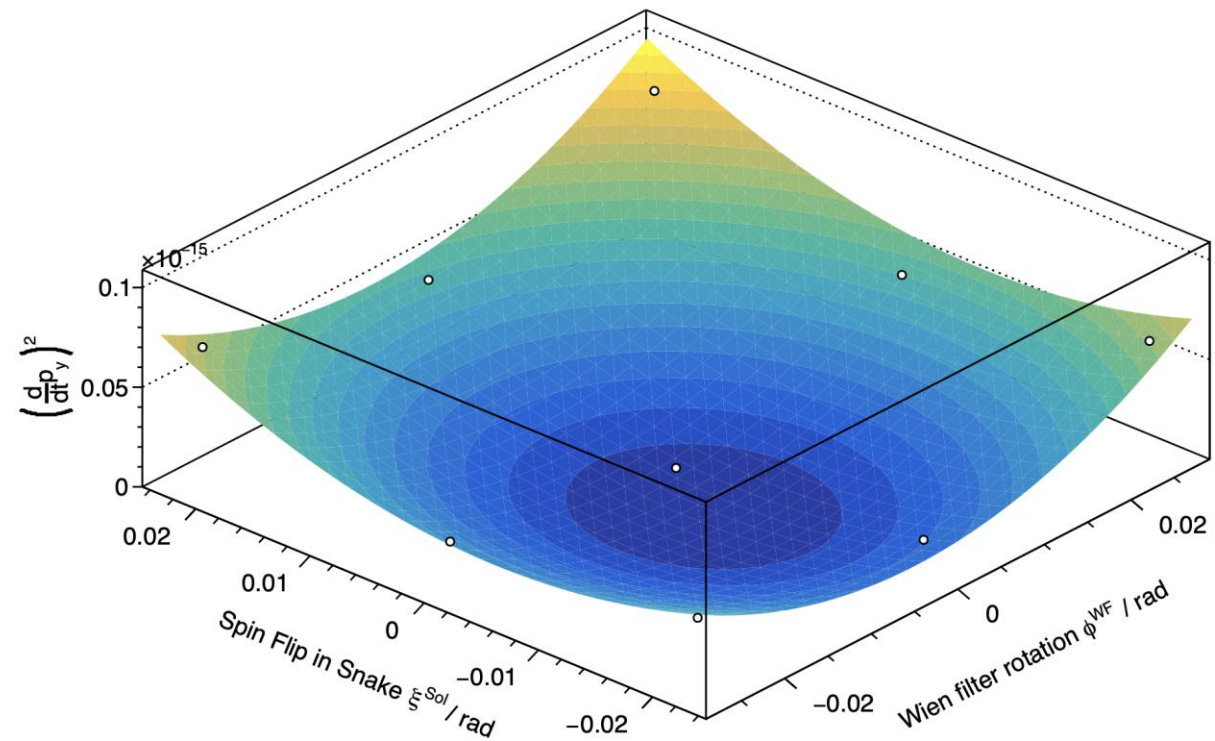
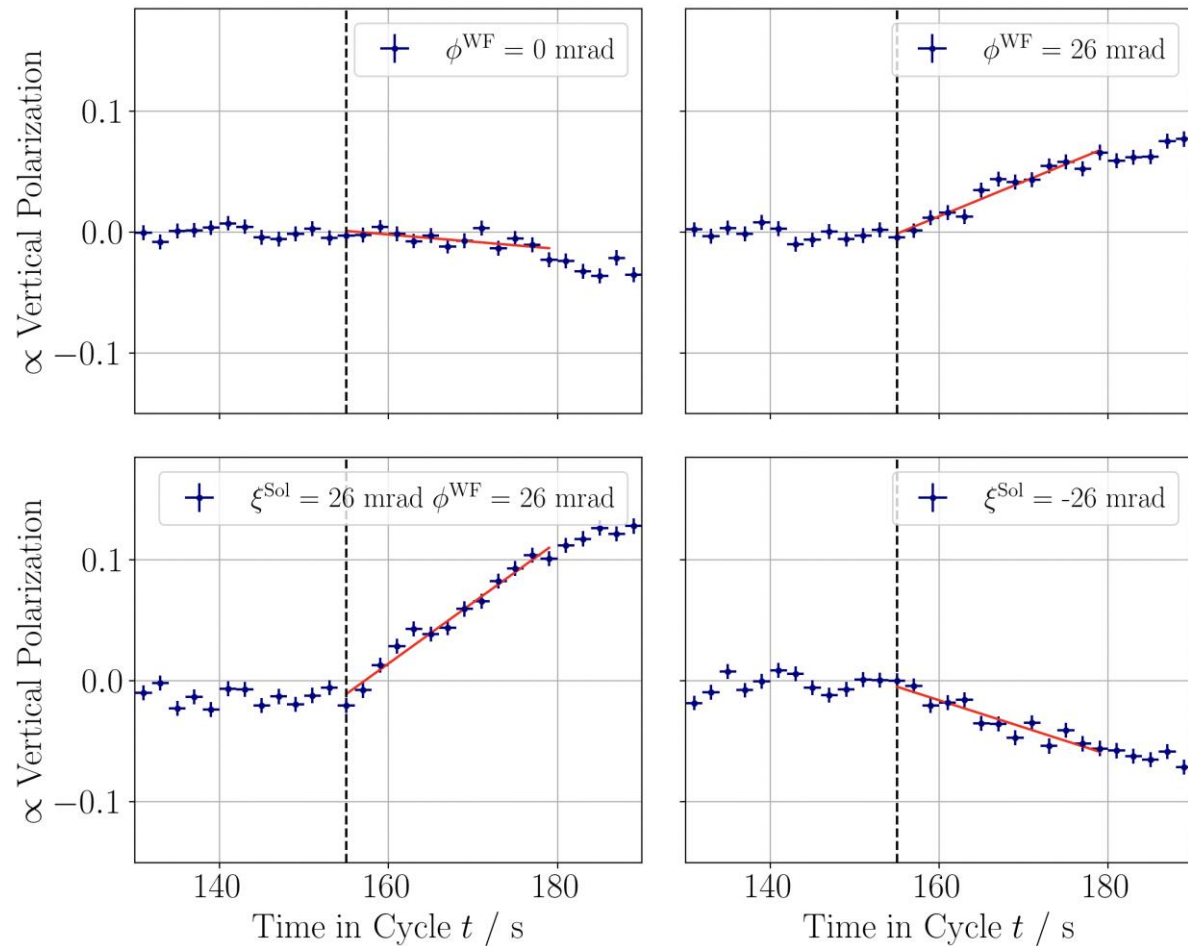
MEASUREMENT PRINCIPLE



Build up rate $\frac{d}{dt} p_y(t)$

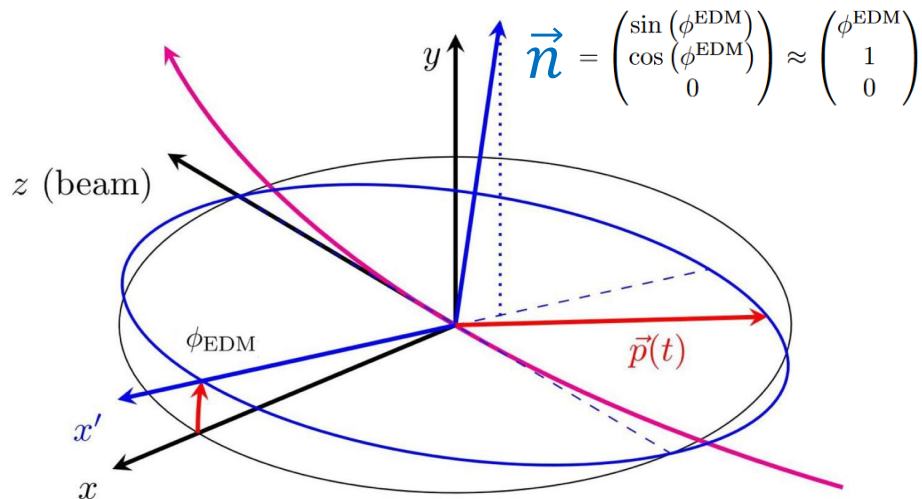
$$\left(\frac{d}{dt} p_y\right)^2 \propto |\vec{n} \times \vec{m}| \propto (\textcolor{red}{n}_x - \textcolor{blue}{\phi}^{\text{WF}})^2 + \dots$$

MEASUREMENT PRINCIPLE



$$\left(\frac{d}{dt}p_y\right)^2 \propto |\vec{n} \times \vec{m}| \propto \left[(\textcolor{red}{n}_x - \phi^{\text{WF}})^2 + (\textcolor{red}{n}_z - \xi^{\text{Sol}})^2\right]$$

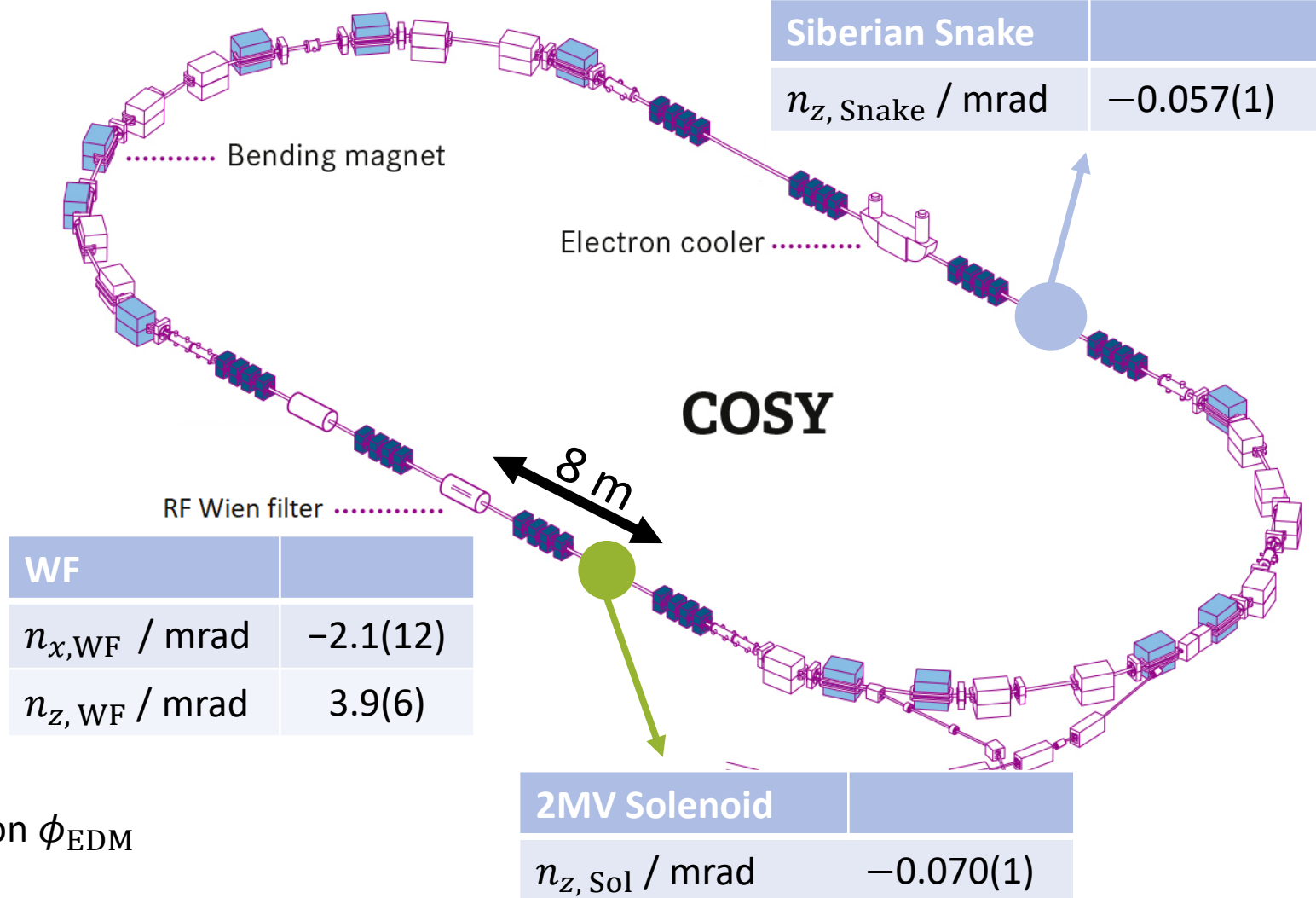
RESULTS



- $n_{z, \text{WF}}$ corresponds to the systematic error on ϕ_{EDM}

- $\sigma_{\text{sys}} = \sigma_{z, \text{WF}}$

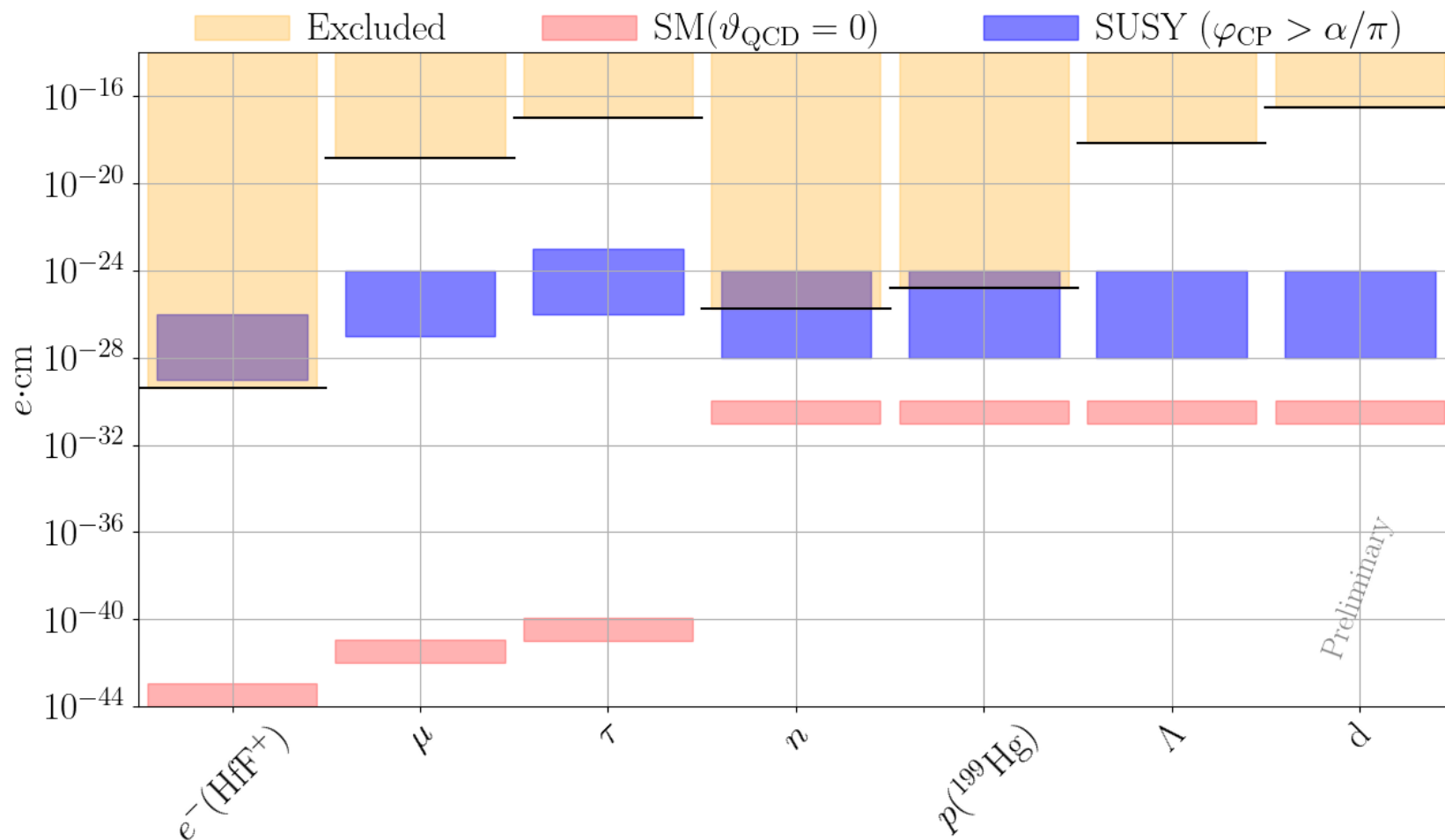
- $d = n_x \frac{G}{2\beta} \frac{q\hbar}{2mc} \rightarrow |d_{\text{p}}^d| < 2.5 \times 10^{-17} e \cdot \text{cm}$ (95% C.L.)



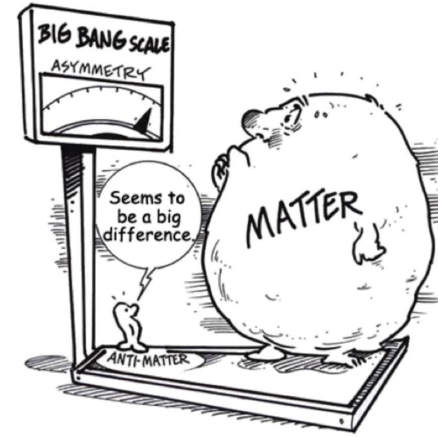
PERMANENT EDM

$$d \propto G$$

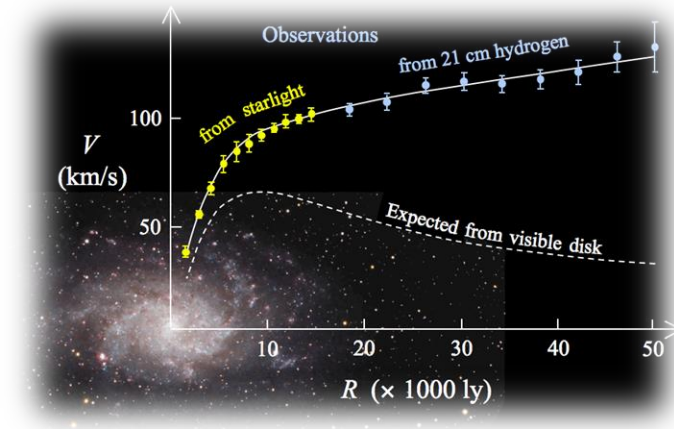
- $G_d \approx -0.143$ vs. $G_\mu \approx 0.001$
- $|d_P^d| < 2.5 \times 10^{-17} e \cdot \text{cm}$
- $|d_P^\mu| < 1.8 \times 10^{-19} e \cdot \text{cm}$ [1]
- COSY was never built for such experiments
- Shutdown of COSY in 2023
- **Future:** Dedicated EDM Rings using $\bar{u}$ and u beams



SUMMARY



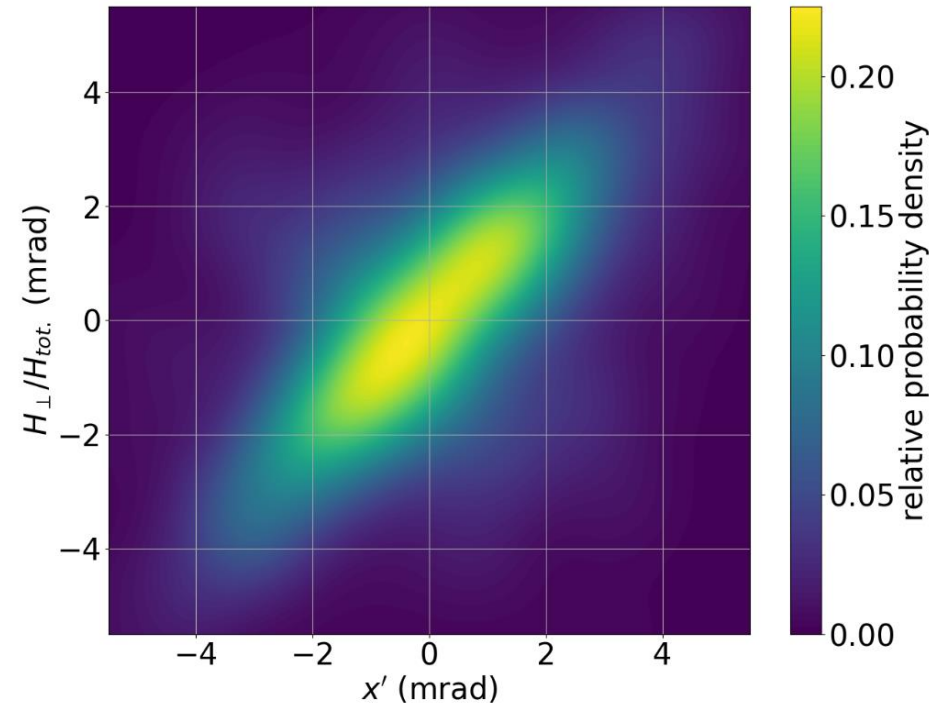
- **Motivation:** EDMs can potentially solve a lot of problems!
- **Goal:** Measure the **influence** of the deuteron EDM on the **beam polarization**
- $\vec{d} = d \frac{\vec{s}}{|\vec{s}|}$ with $d = d_P + d_{\text{Osc}} \cos(\omega_a + \phi_a)$ and $\omega_a = \frac{m_a c^2}{\hbar}$
- $|d_P| < 2.5 \times 10^{-17} e \cdot \text{cm}$ // Preprint on [Arxiv](#) 2602.20828 (2026)
- $|d_{\text{Osc}}| < 6.4 \times 10^{-23} e \cdot \text{cm}$ // [PRX](#) 13, 031004 (2023)



$$L_{\bar{\theta}_{\text{QCD}}} = -\bar{\theta}_{\text{QCD}} \frac{g_s^2}{64\pi^2} \varepsilon^{\mu\nu\alpha\beta} G_{\mu\nu}^a G_{\alpha\beta}^a$$

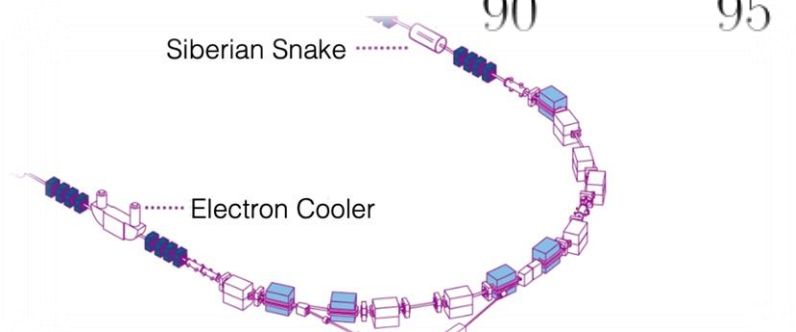
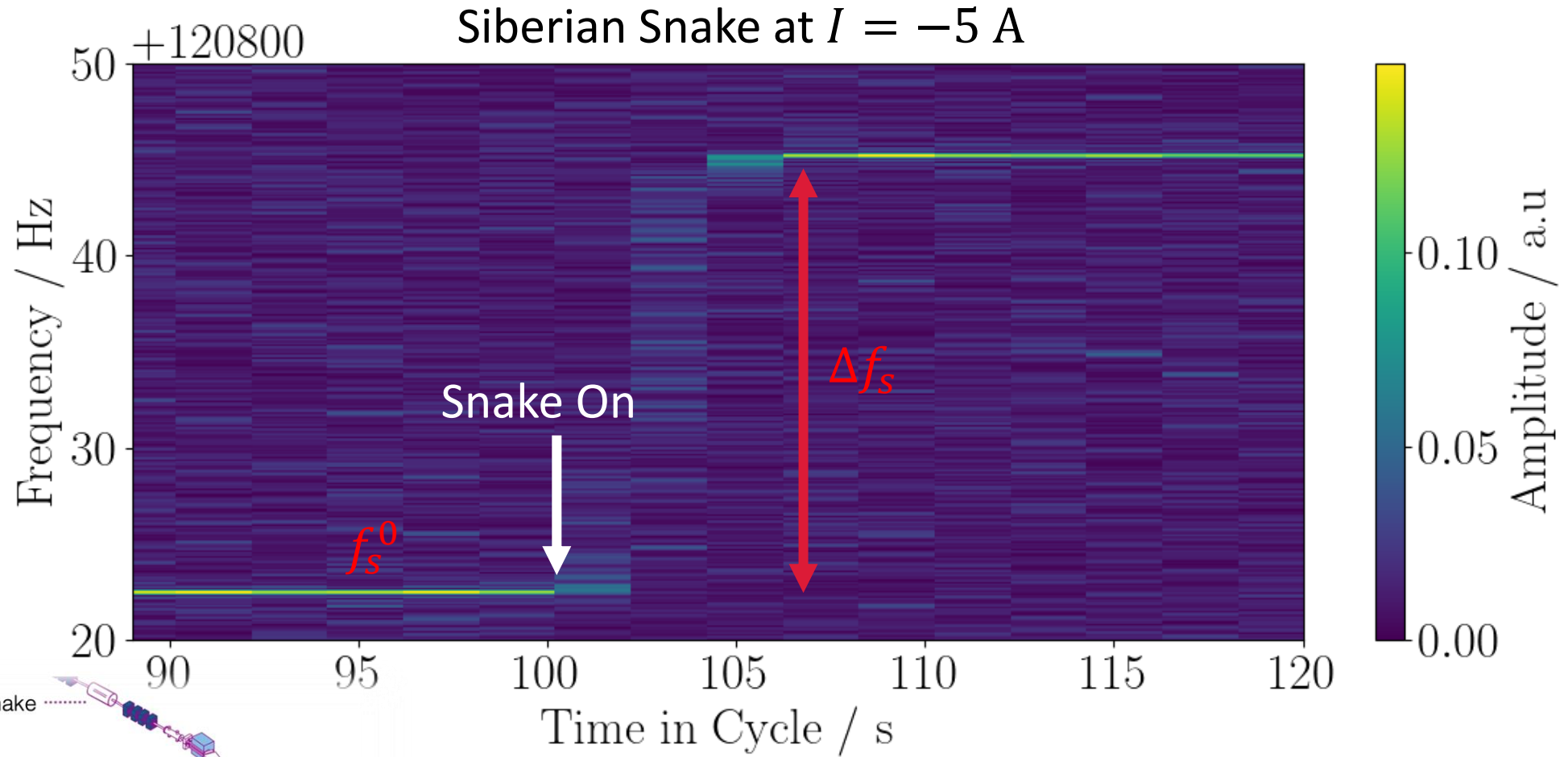
BACK UP

PERMANENT DEUTERON EDM SEARCH



INVARIANT SPIN AXIS AT THE SOLENOIDS

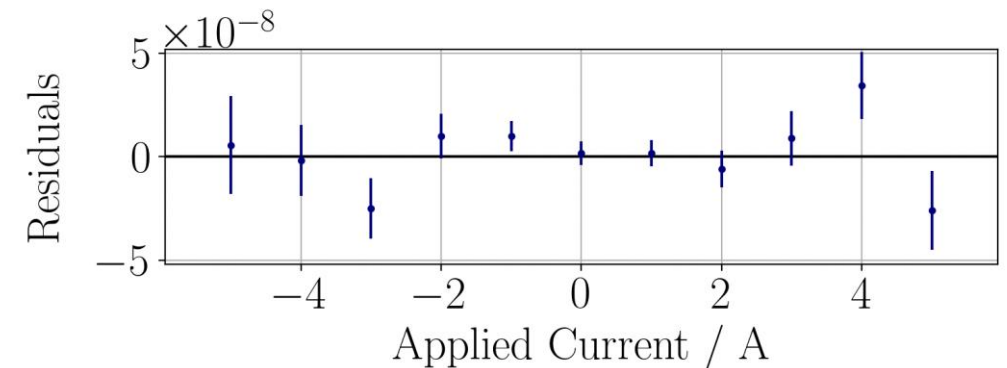
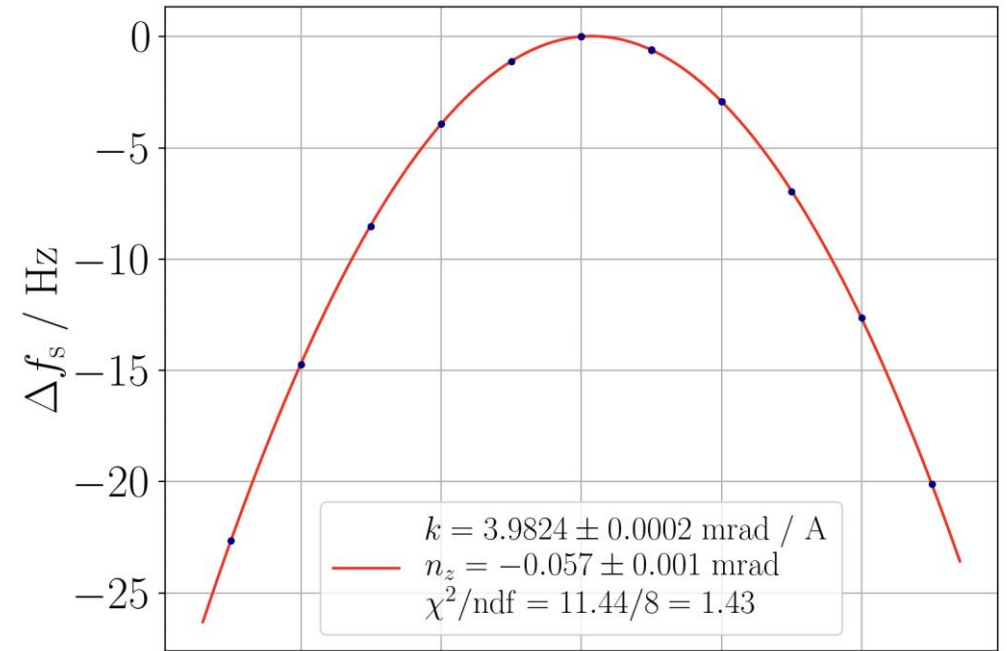
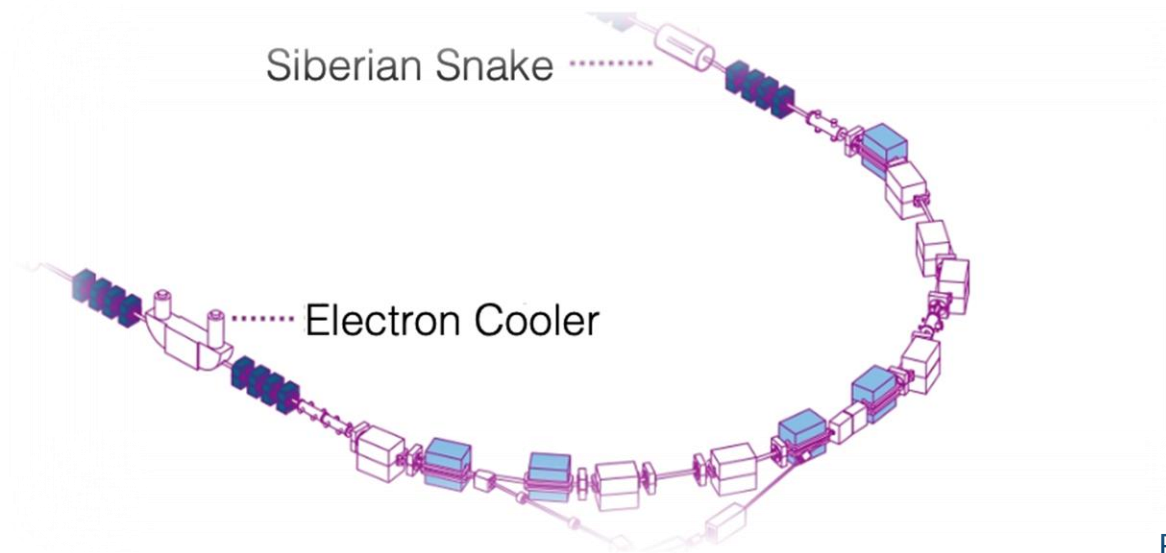
Methodology



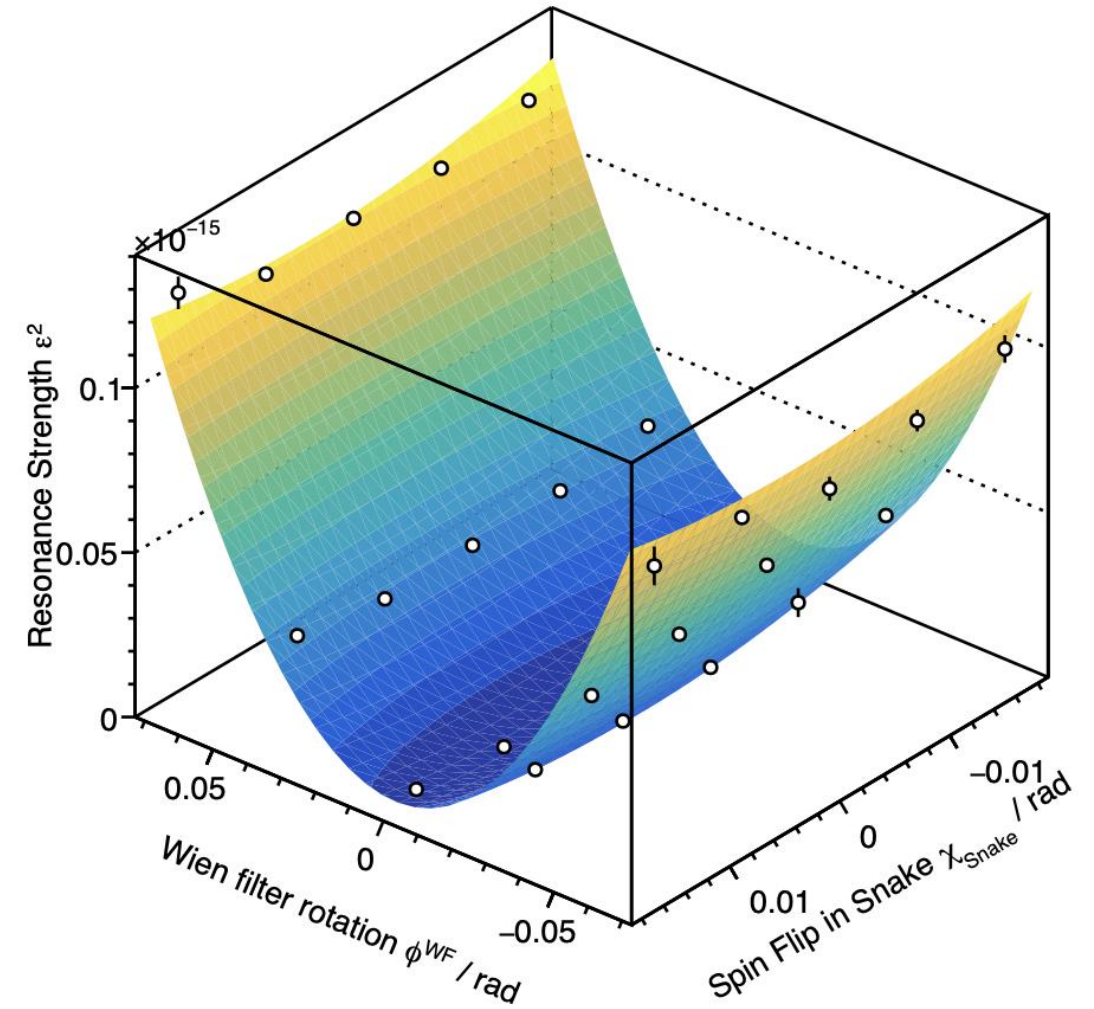
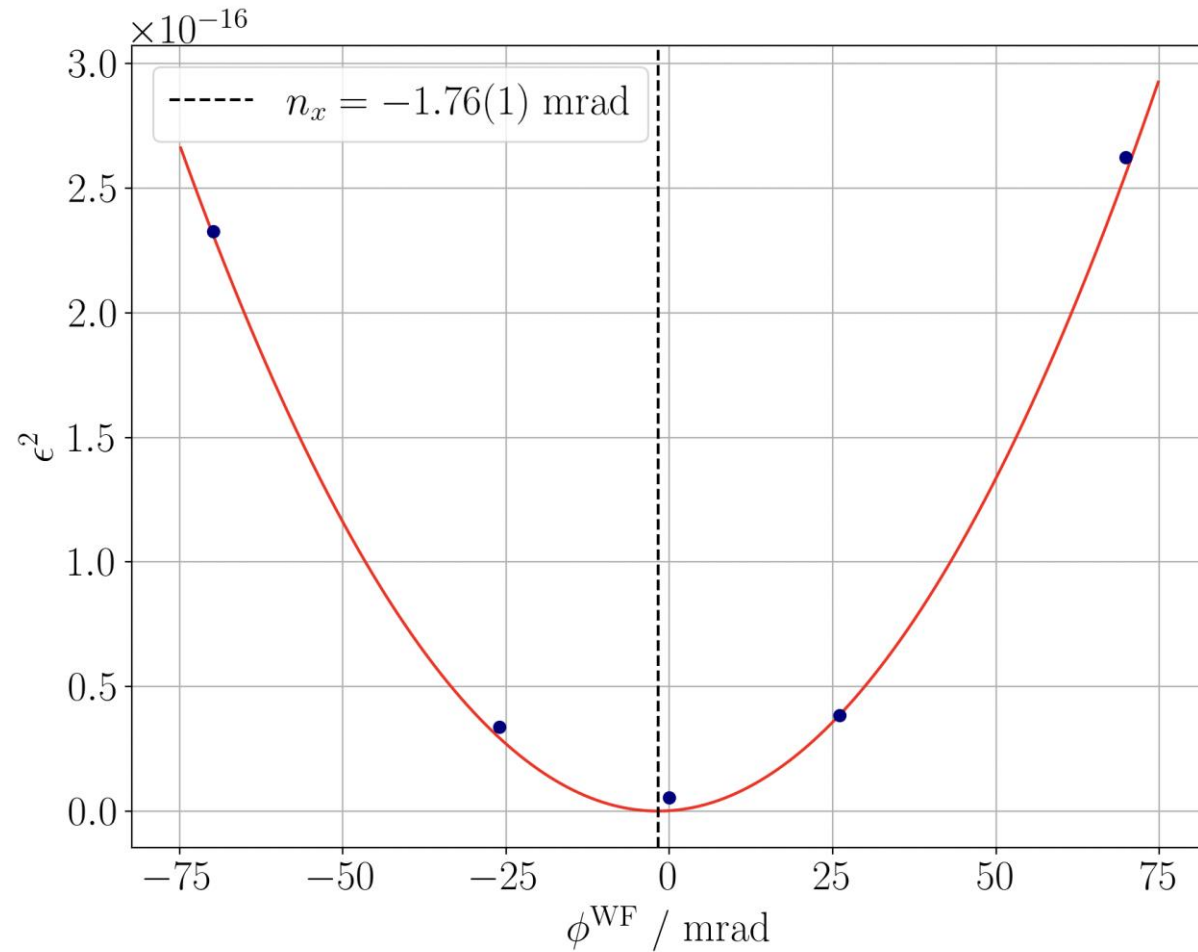
INVARIANT SPIN AXIS AT THE SOLENOIDS

$$\Delta f_s = \frac{f_{\text{rev}}}{4\pi} \left[\frac{\cos\left(\pi \frac{f_s^0}{f_{\text{rev}}}\right)}{2} k^2 I^2 + n_z k I \right]$$

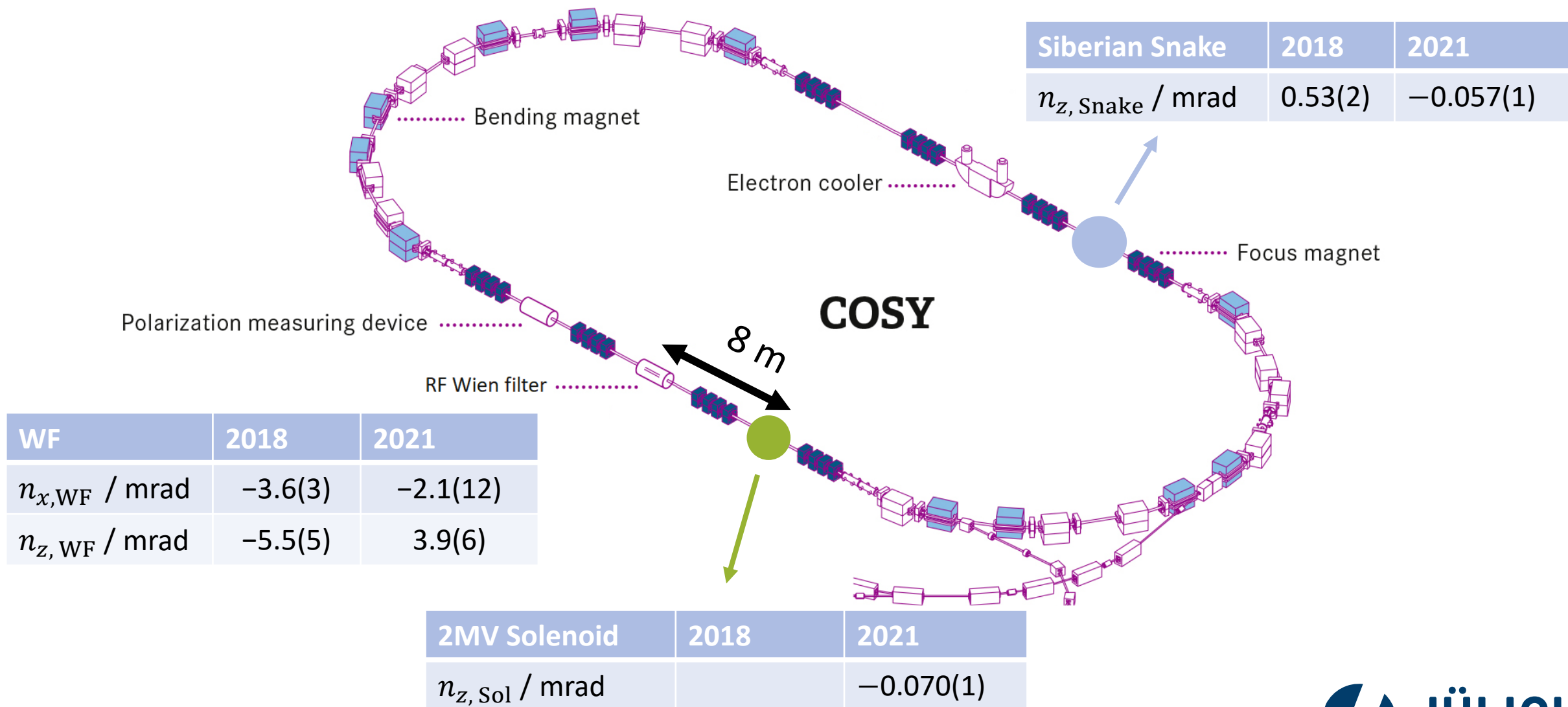
- k translates the coil current into a spin tilt angle
- n_z denotes the z component of the invariant spin axis



DETAILED LOOK INTO DATA



RESULTS



ALIGNMENT OF THE EDM WITH THE SPIN VECTOR

Elementary quantum mechanics of the neutron with an electric dipole moment

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Contributed by Gordon Baym, May 16, 2016 (sent for review February 16, 2016; reviewed by Wick C. Haxton and Thomas Schaefer)

- $\vec{d}$ has zero expectation value for states of a given parity
 - Operator is odd under parity transformation $\vec{d} = -\mathcal{P}^\dagger \vec{d} \mathcal{P}$ $\longrightarrow \langle a | \vec{d} | a \rangle = -\langle a | \vec{d} | a \rangle = 0$
- Allow for a parity violating contribution η

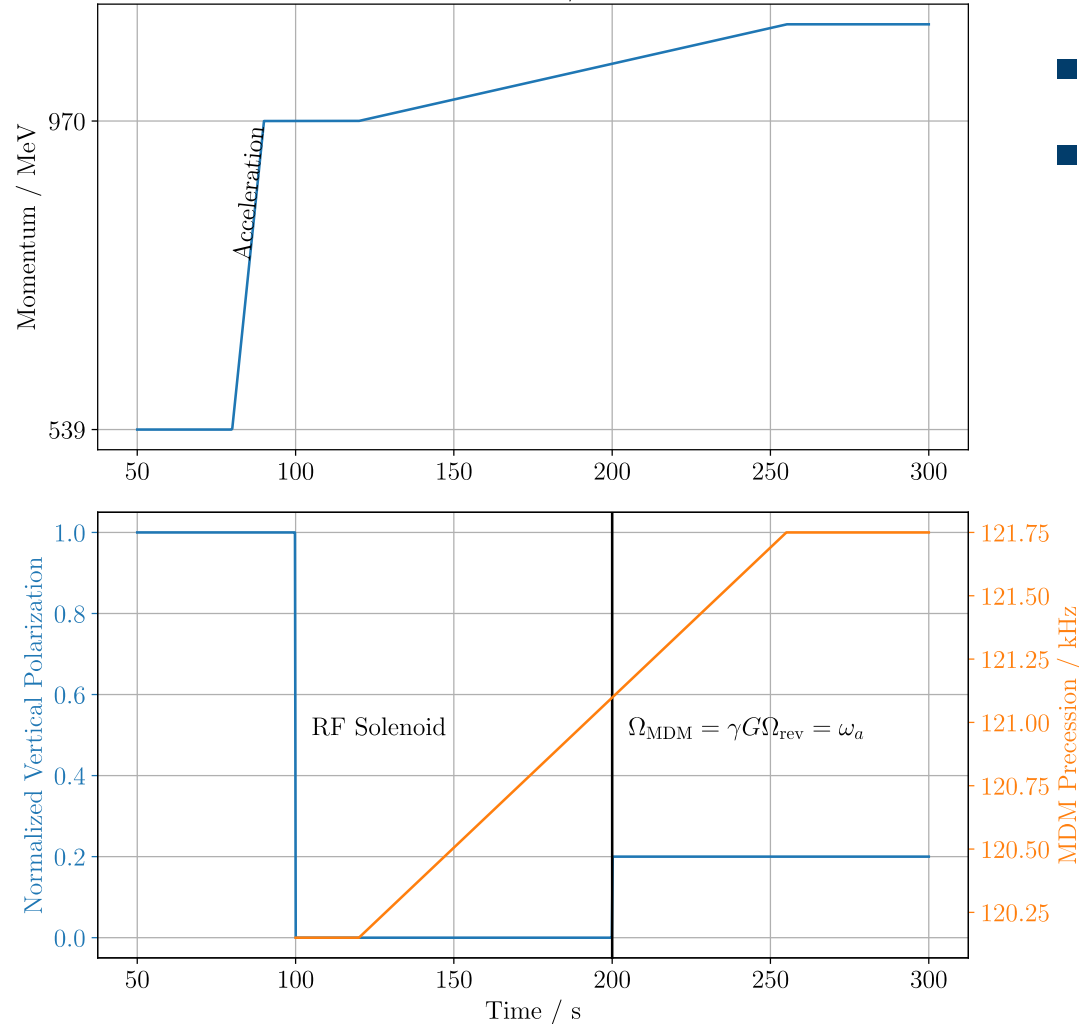
$$|a\rangle = |+\rangle + \eta |-\rangle \longrightarrow \langle a | \vec{d} | a \rangle = \eta \langle + | \vec{d} | - \rangle + \text{c.c.} \neq 0.$$

- A proof that EDM and spins are aligned is given in the paper above $\langle a | \vec{d} | a \rangle \propto \langle a | \vec{S} | a \rangle$

AXION SEARCH

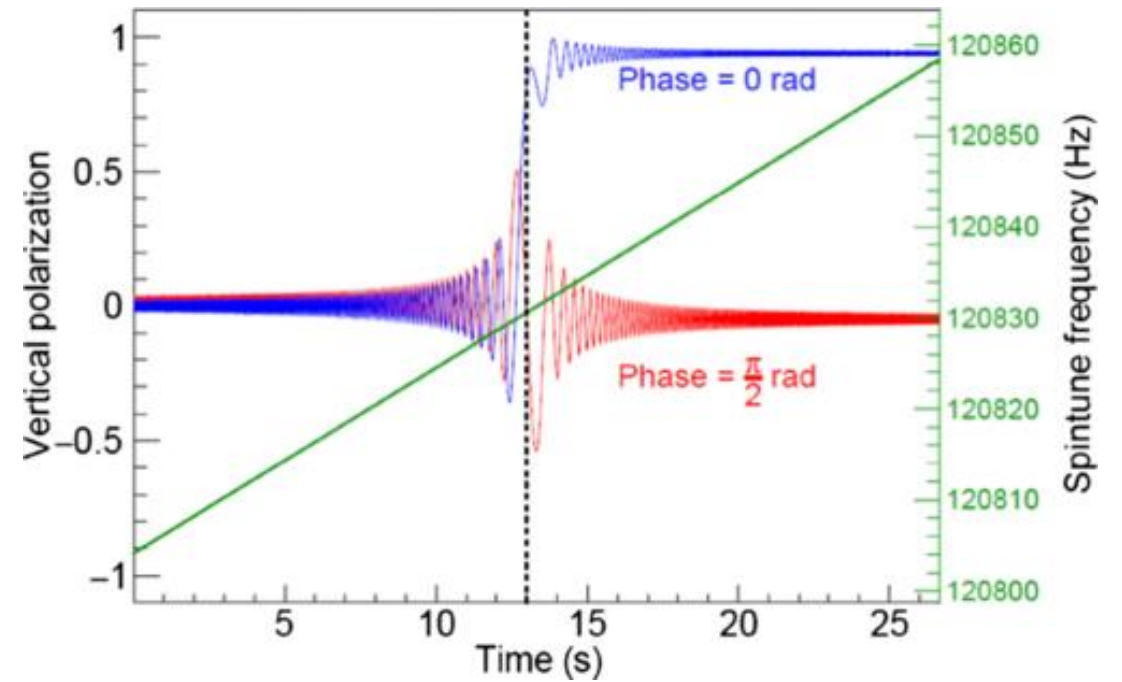
s. Karanth (PhD Work)

Measurement Principle / Expected Signal



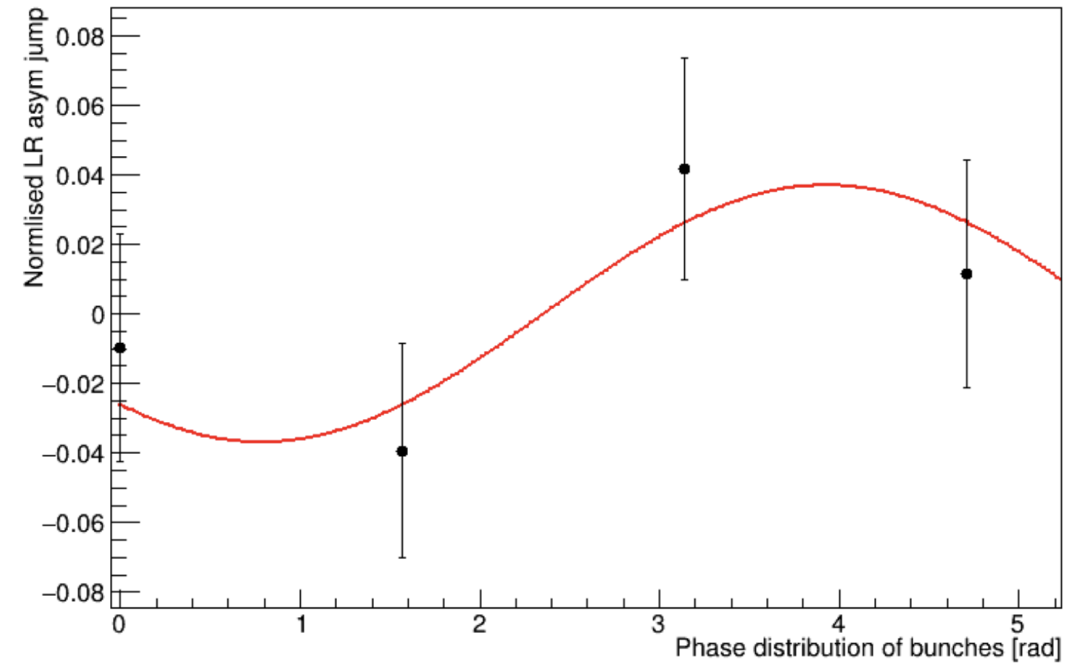
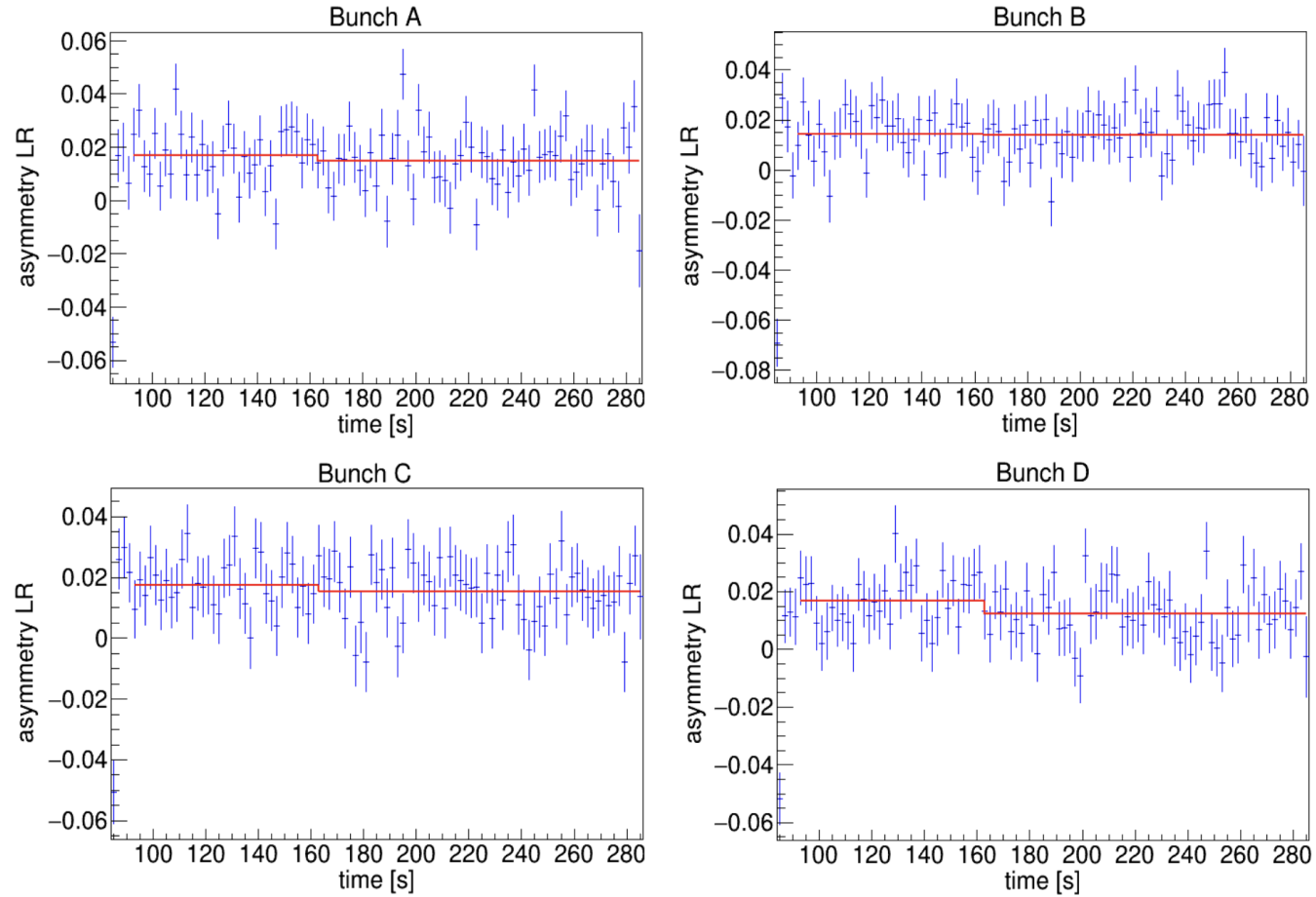
$$d = d_{\text{DC}} + d_{\text{AC}}^d \cos(\omega_a + \phi_a)$$

- Problem: Phase is unknown!
- Solution: Inject 4 bunches with different spin directionality!



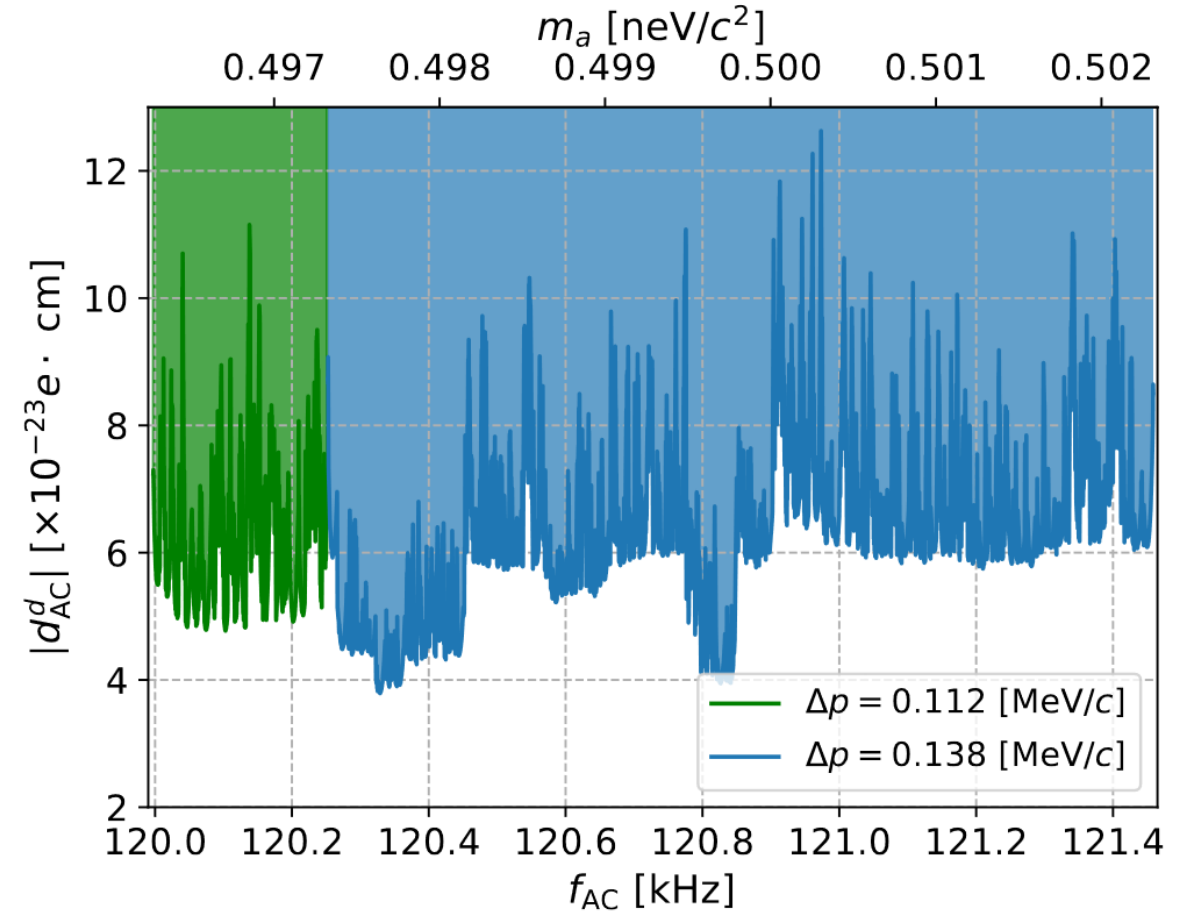
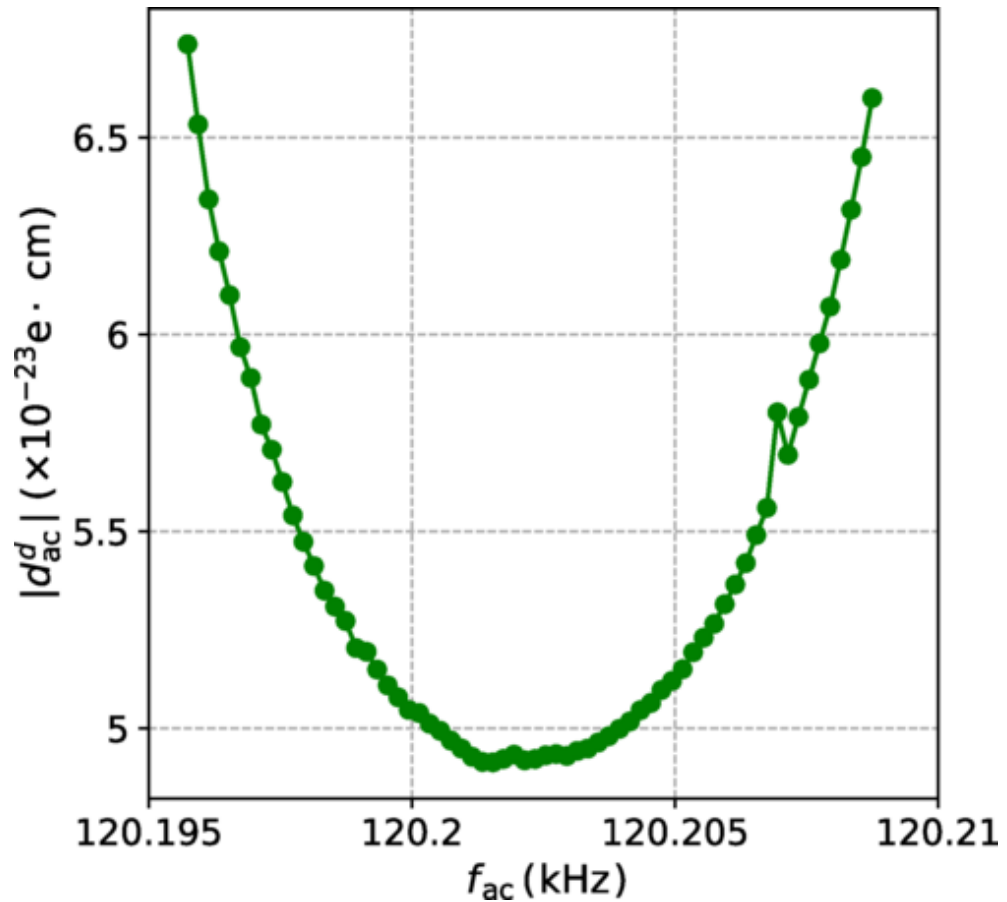
AXION SEARCH

s. Karanth (PhD Work)



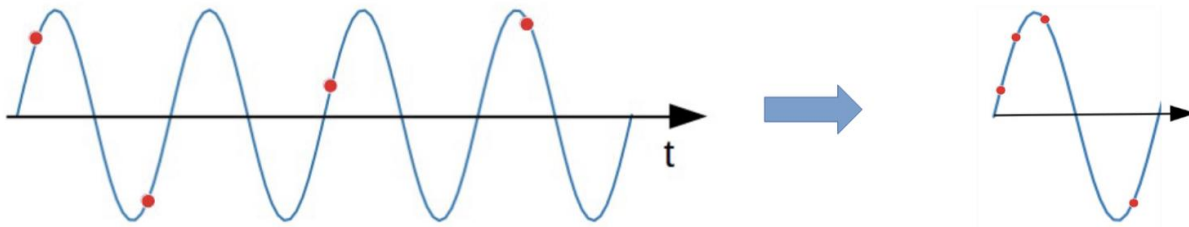
AXION SEARCH

S. Karanth (PhD Work)



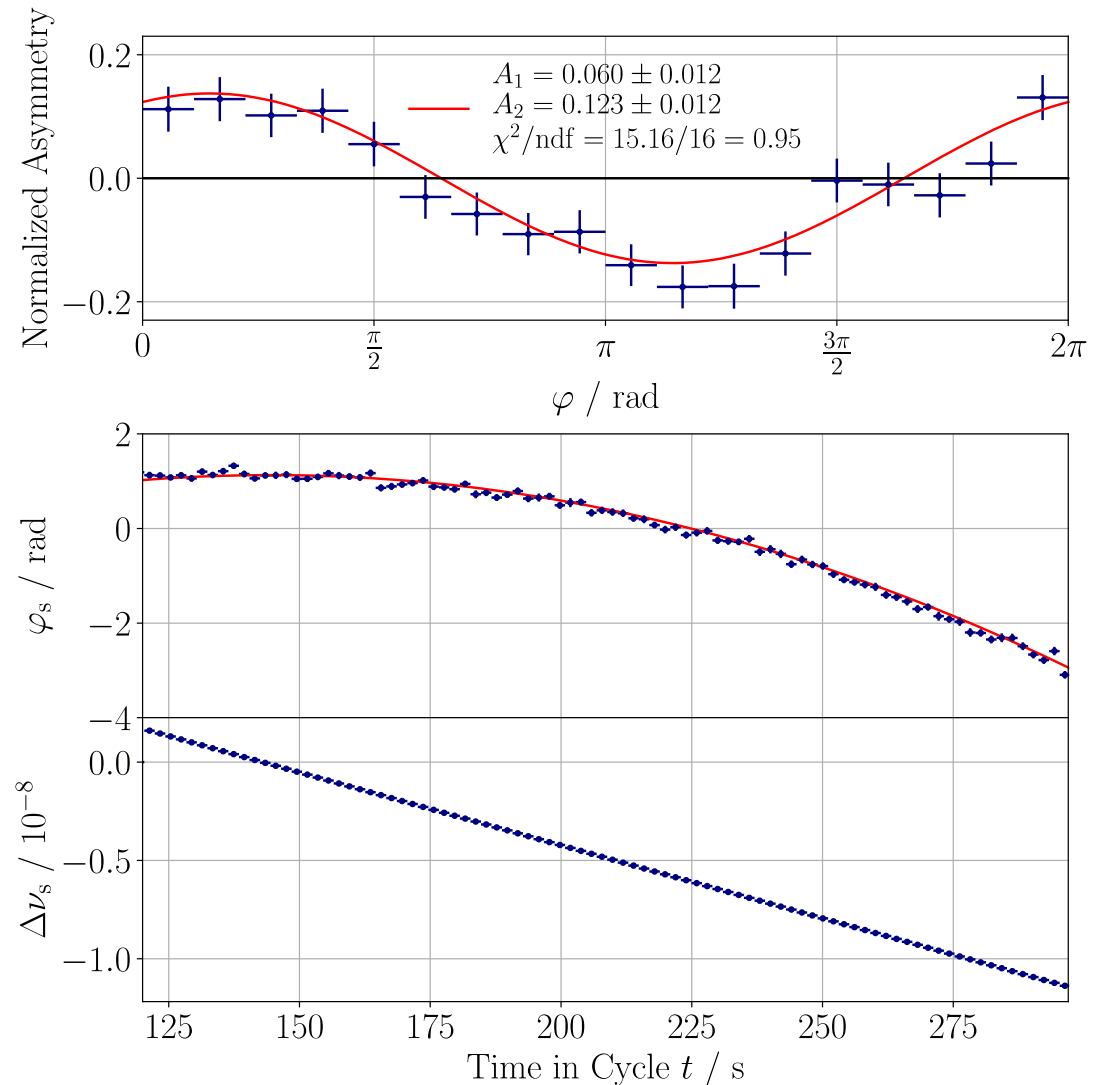
SPIN TUNE MEASUREMENT

- Spin precesses with $f_s = \gamma G f_{\text{rev}} \approx 121 \text{ kHz}$
- Detector handles event rates $15000 \frac{1}{\text{s}}$
 - 1 hit per 10 precessions
 - No direct fit possible
- Assume a fixed frequency ν_s^{fixed}



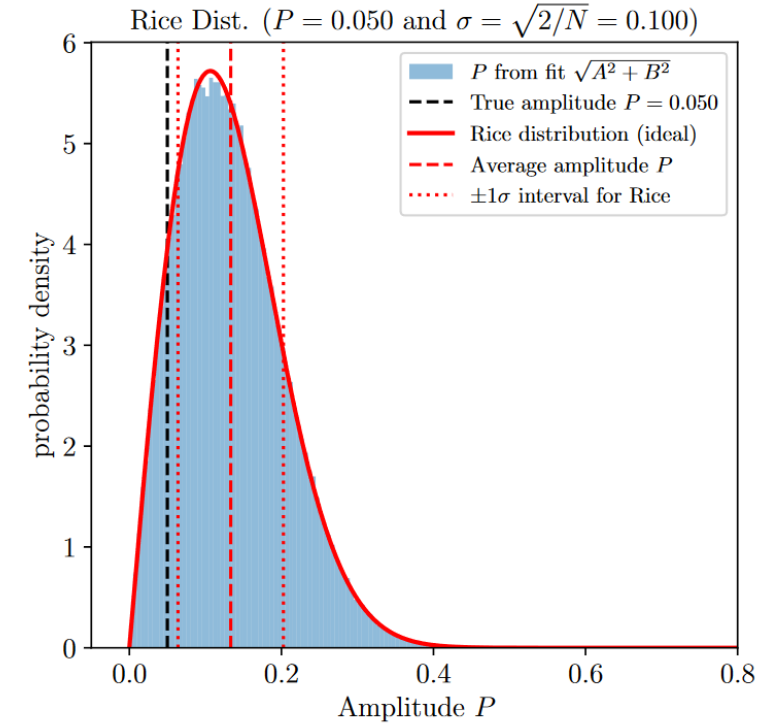
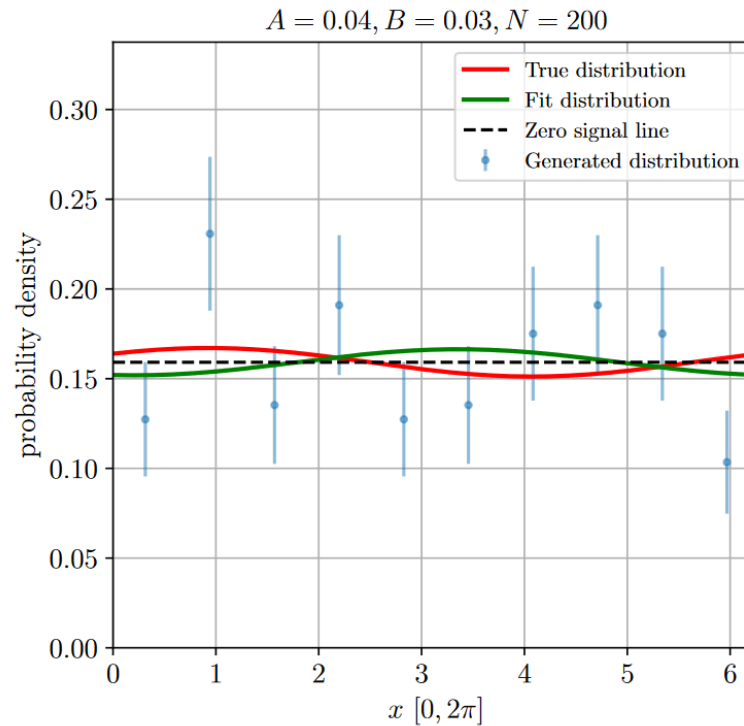
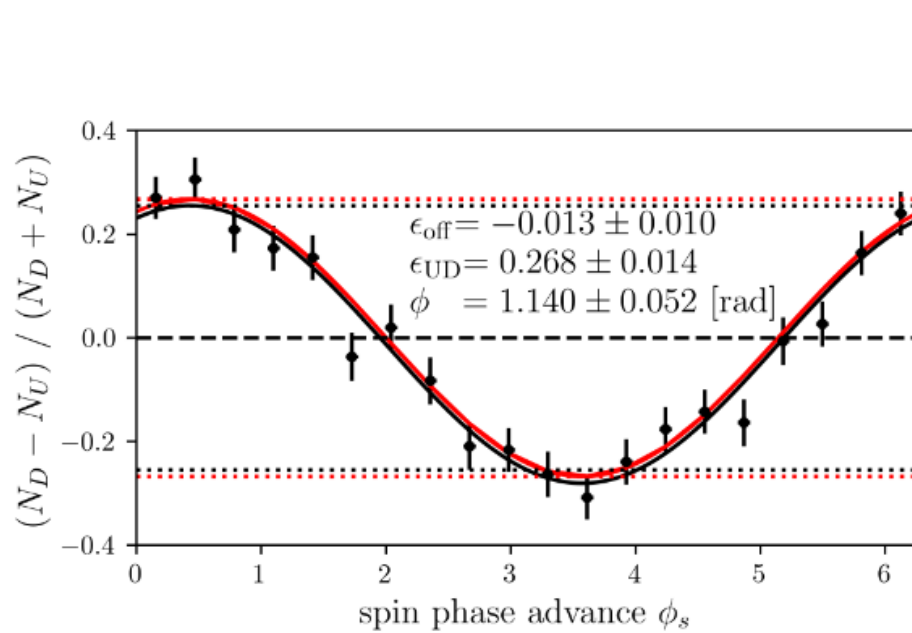
- The change of spin tune is given by

$$\nu_s(t) = \nu_s^{\text{fixed}} + \frac{1}{2\pi f_{\text{rev}}} \frac{d\varphi_s}{dt} = \nu_s^{\text{fixed}} + \Delta\nu_s(t)$$



BIAS IN FITTING PROCEDURE

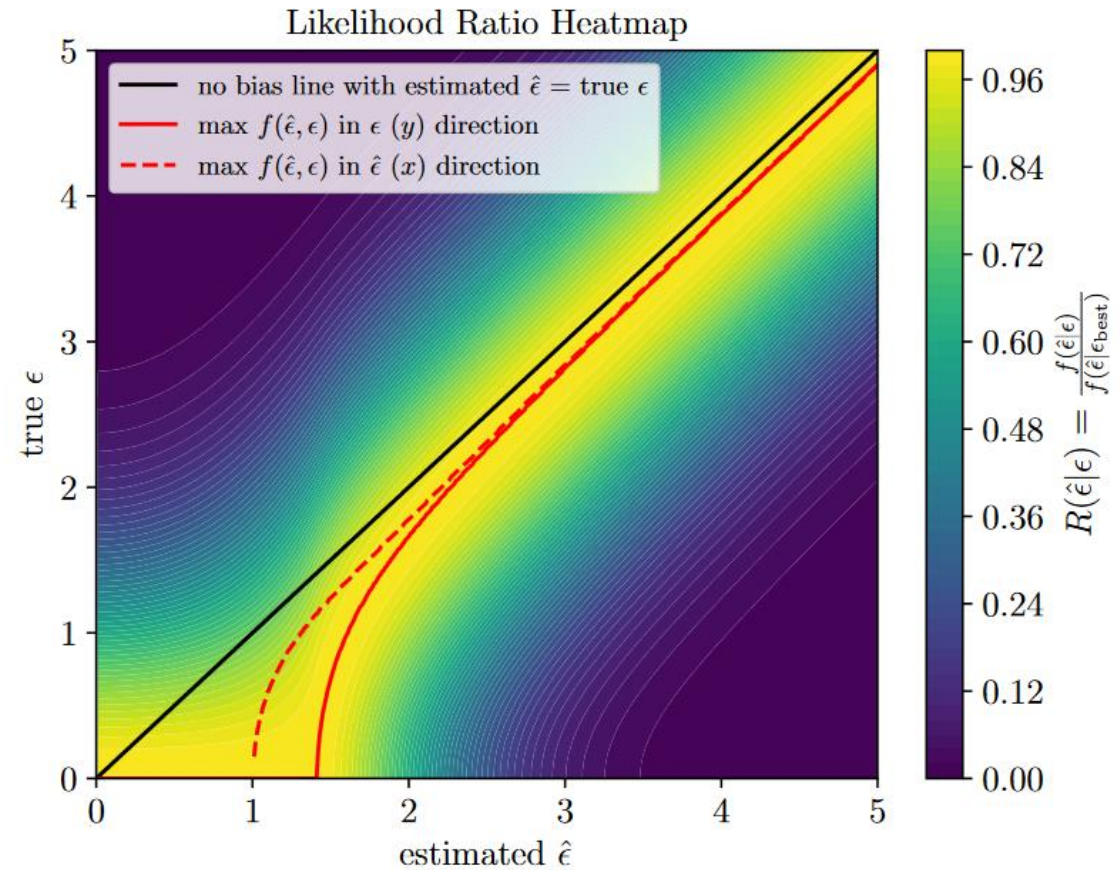
V. Tempel (Master Thesis)



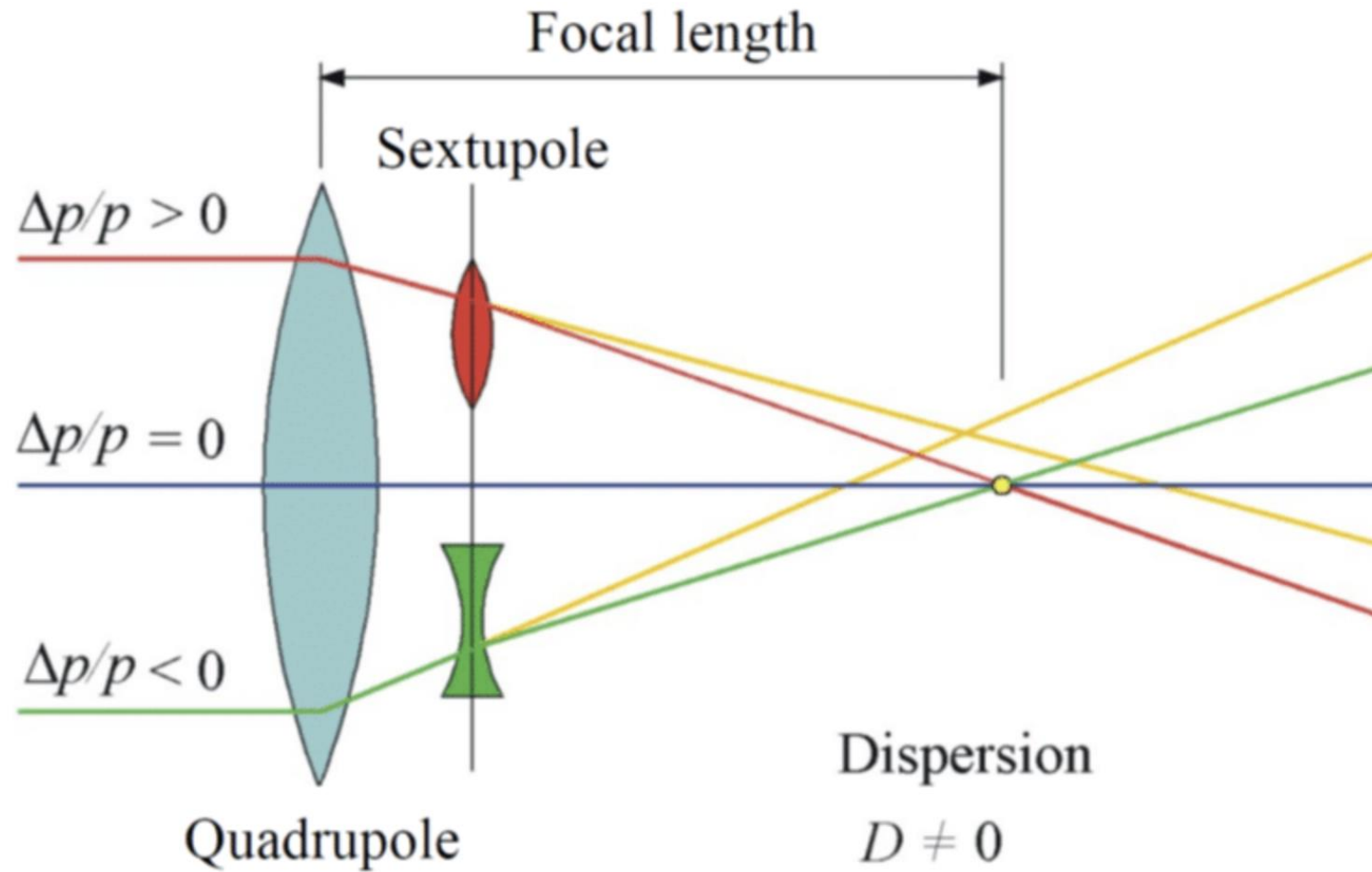
BIAS IN FITTING PROCEDURE

V. Tempel (Master Thesis)

- Confidence Intervals can be properly constructed using Feldman – Cousin Algorithms
- However: Low Polarization Events are not really relevant for the experiment

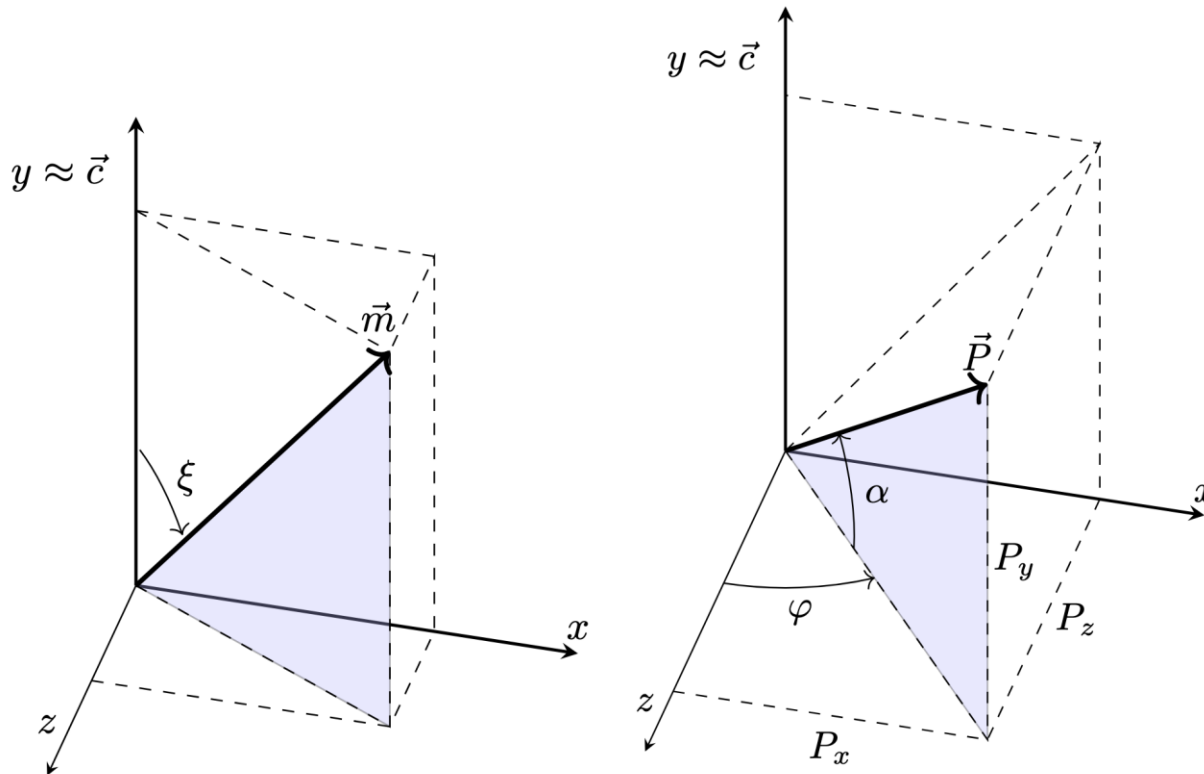


SEXTUPOLE CORRECTION FOR SCT ENHANCEMENT



PHASE FEEDBACK

- Tasks of the phase feedback:
 - Measures the spin precession frequency and adjusts the Wien filter frequency to it
 - Provides a fixed phase relation between both frequencies



$$\frac{d\alpha}{dn} = \frac{k}{2} \cos \varphi_{\text{rel}}$$

$$\frac{d\varphi_{\text{rel}}}{dn} = \frac{k}{2} (\sin \varphi_{\text{rel}} \tan \alpha + q)$$

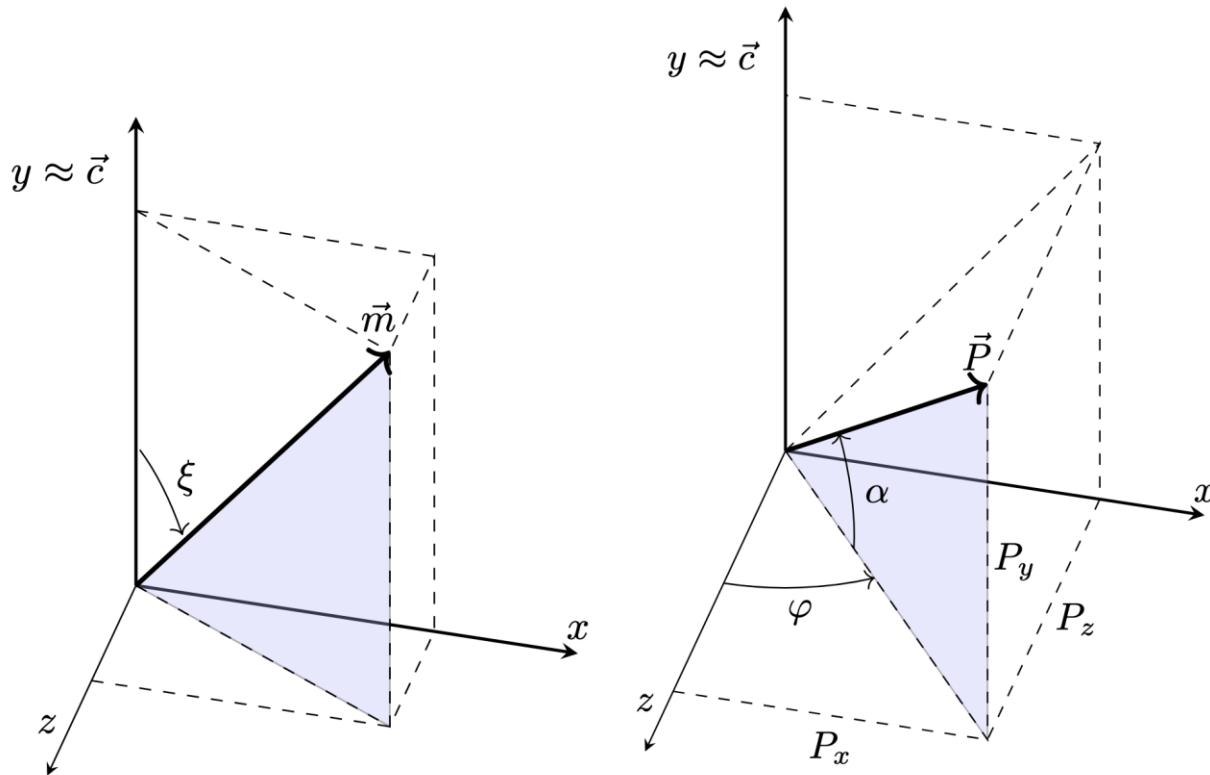
$$k = -\chi_0 \sin \xi = -4\pi\epsilon$$

$$q = \frac{4\pi\Delta f}{kf_{\text{rev}}}$$

$$\epsilon = \frac{f_v}{f_{\text{rev}}} = \frac{1}{4\pi} \chi_0 |\vec{c} \times \vec{m}| = \frac{1}{4\pi} \chi_0 \sin \xi$$

PHASE FEEDBACK

Slope Method



- Phase Feedback prevents the relative phase from changing

$$\frac{d\alpha}{dn} = \frac{k}{2} \cos \varphi_{\text{rel}}$$

$$\frac{d\varphi_{\text{rel}}}{dn} = \frac{k}{2} (\sin \varphi_{\text{rel}} \tan \alpha + q) \longrightarrow \frac{d\varphi_{\text{rel}}}{dn} = 0$$

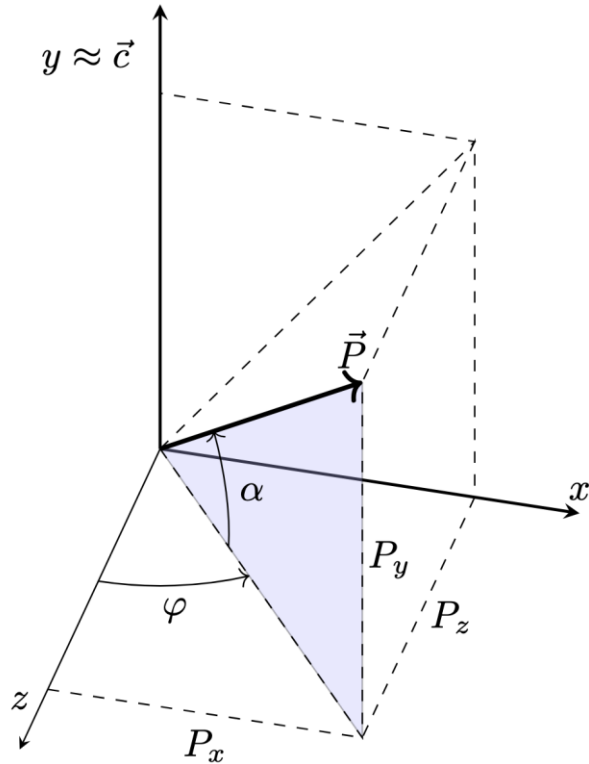
$$\frac{d\alpha}{dt} = 2\pi\epsilon f_{\text{rev}} \cos \varphi_{\text{rel}}$$

$$\epsilon = \frac{f_v}{f_{\text{rev}}} = \frac{1}{4\pi} \chi_0 |\vec{c} \times \vec{m}| = \frac{1}{4\pi} \chi_0 \sin \xi$$

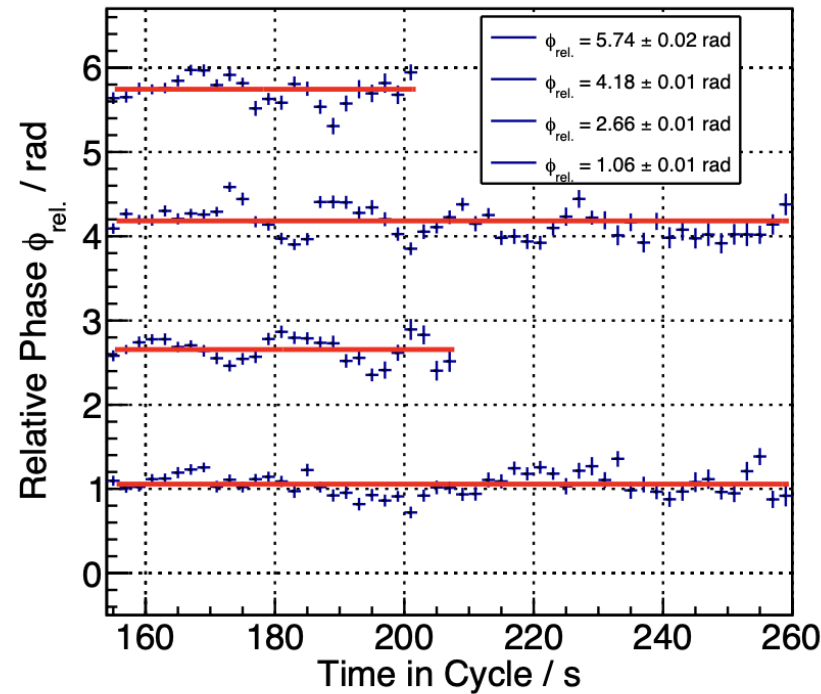
PHASE FEEDBACK

Slope Method

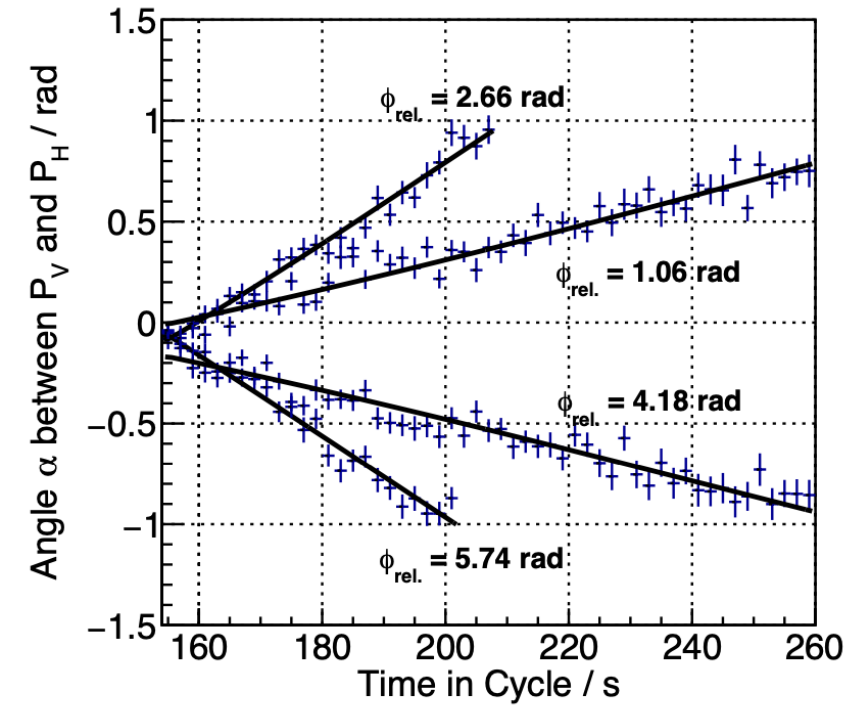
$$\epsilon = \frac{f_v}{f_{\text{rev}}} = \frac{1}{4\pi} \chi_0 |\vec{c} \times \vec{m}| = \frac{1}{4\pi} \chi_0 \sin \xi$$



$$\frac{d\varphi_{\text{rel}}}{dn} = 0$$

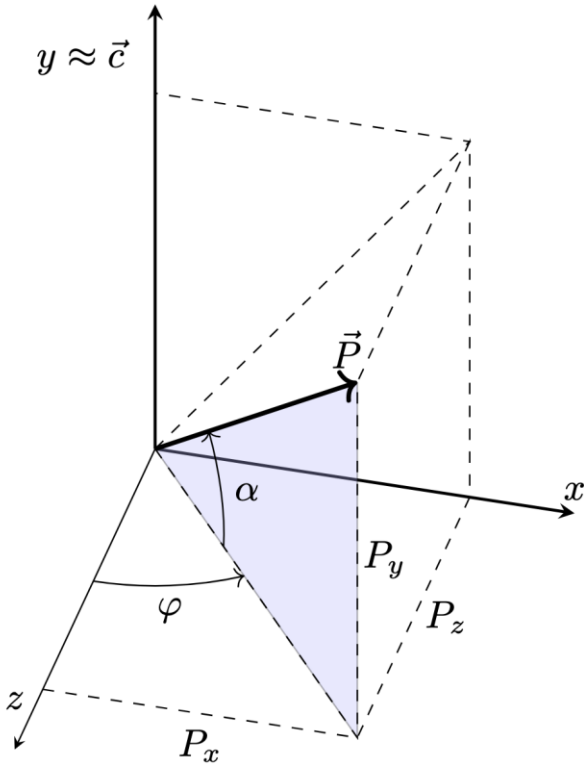


$$\frac{d\alpha}{dt} = 2\pi\epsilon f_{\text{rev}} \cos \varphi_{\text{rel}}$$



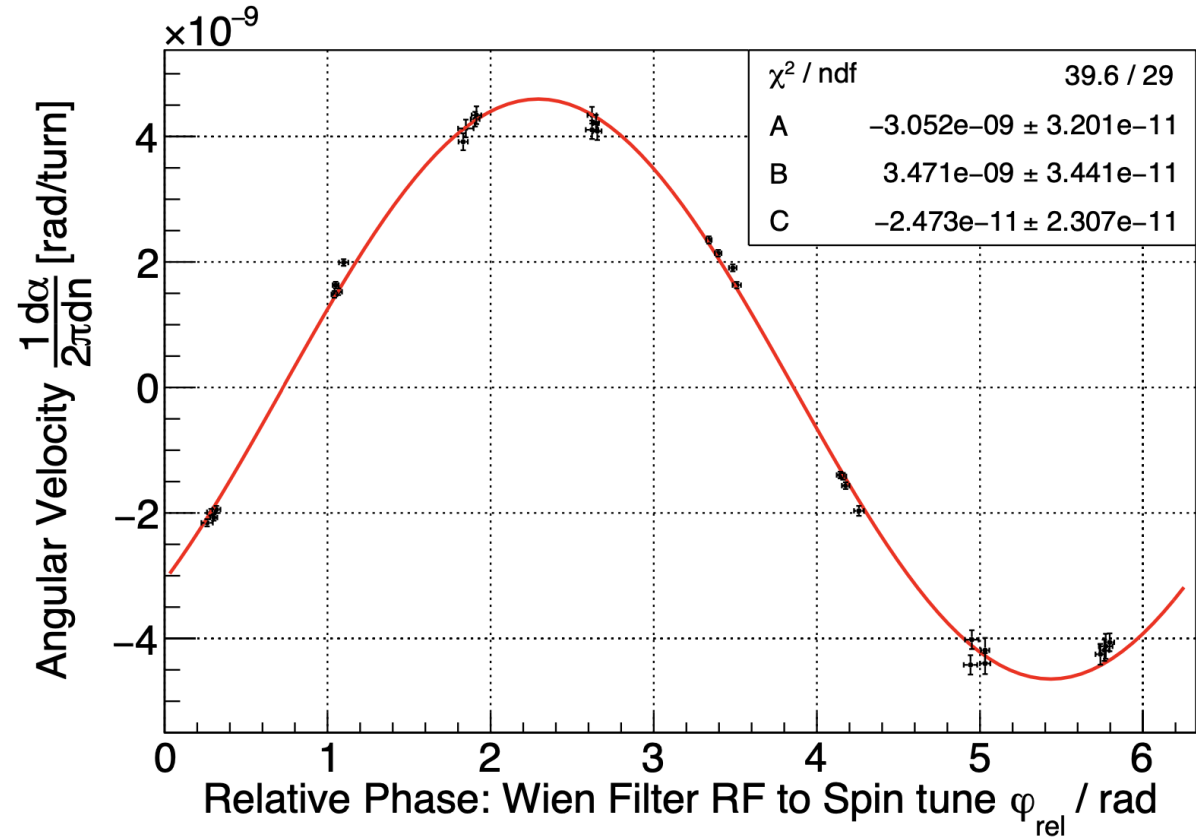
PHASE FEEDBACK

Slope Method



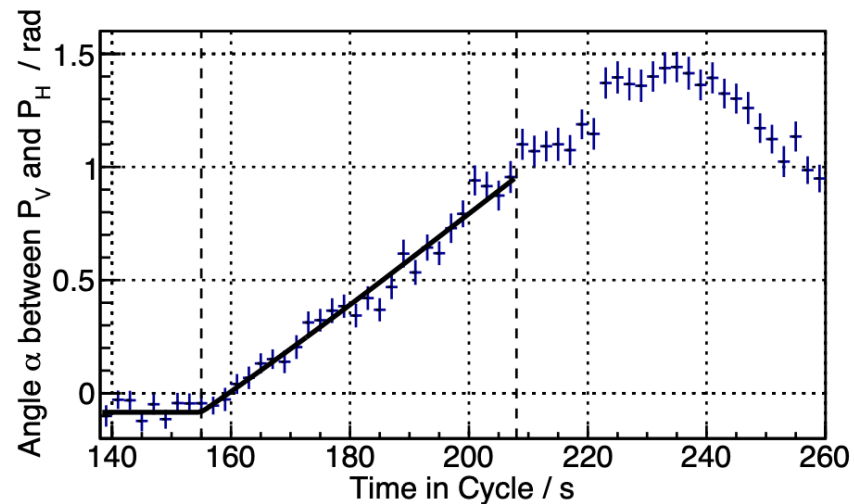
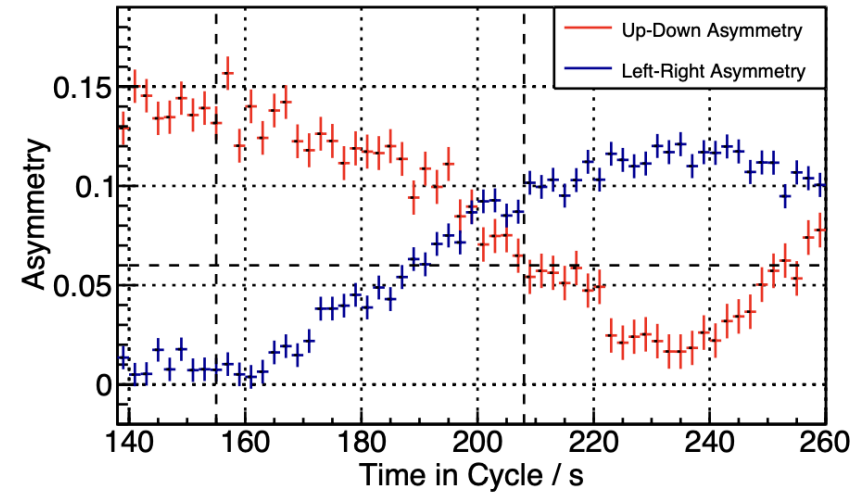
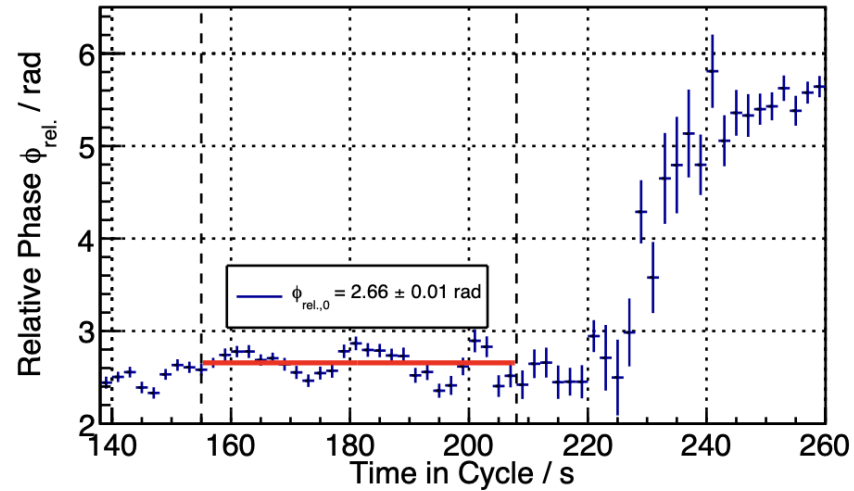
$$\epsilon = \frac{f_v}{f_{\text{rev}}} = \frac{1}{4\pi} \chi_0 |\vec{c} \times \vec{m}| = \frac{1}{4\pi} \chi_0 \sin \xi$$

$$\frac{d\alpha}{dt} = 2\pi\epsilon f_{\text{rev}} \cos \varphi_{\text{rel}}$$

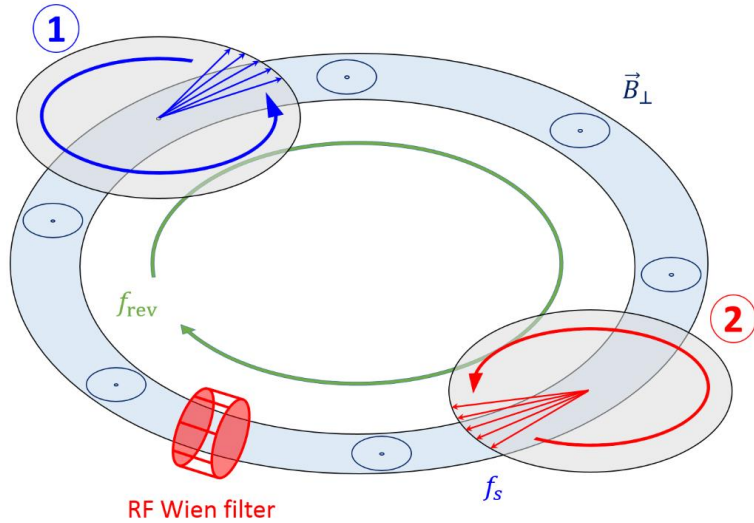


PHASE FEEDBACK

Slope Method - Limitations



PHASE FEEDBACK



Blue Bunch

$$\frac{d\alpha}{dn} = \frac{k}{2} \cos \varphi_{\text{rel}}$$

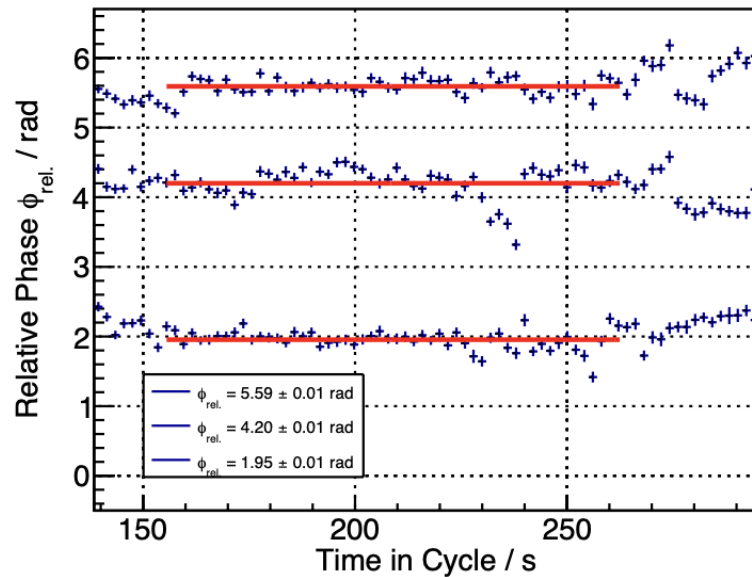
$$\frac{d\varphi_{\text{rel}}}{dn} = \frac{k}{2} (\sin \varphi_{\text{rel}} \tan \alpha + q) \rightarrow \frac{d\varphi_{\text{rel}}}{dn} = 0$$

Red Bunch

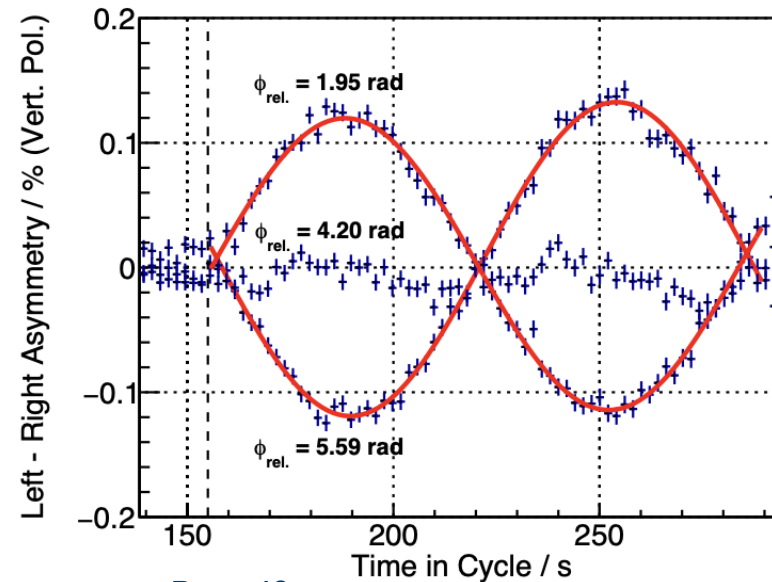
$$\frac{d\alpha}{dn} = \frac{k}{2} \cos \varphi_{\text{rel}}$$

$$\frac{d\varphi_{\text{rel}}}{dn} = \frac{k}{2} (\sin \varphi_{\text{rel}} \tan \alpha + q)$$

$$p_y(t) = \sin \alpha(t) = \cos \varphi_{\text{rel}}(0) \sin(2\pi \epsilon f_{\text{rev}} t).$$

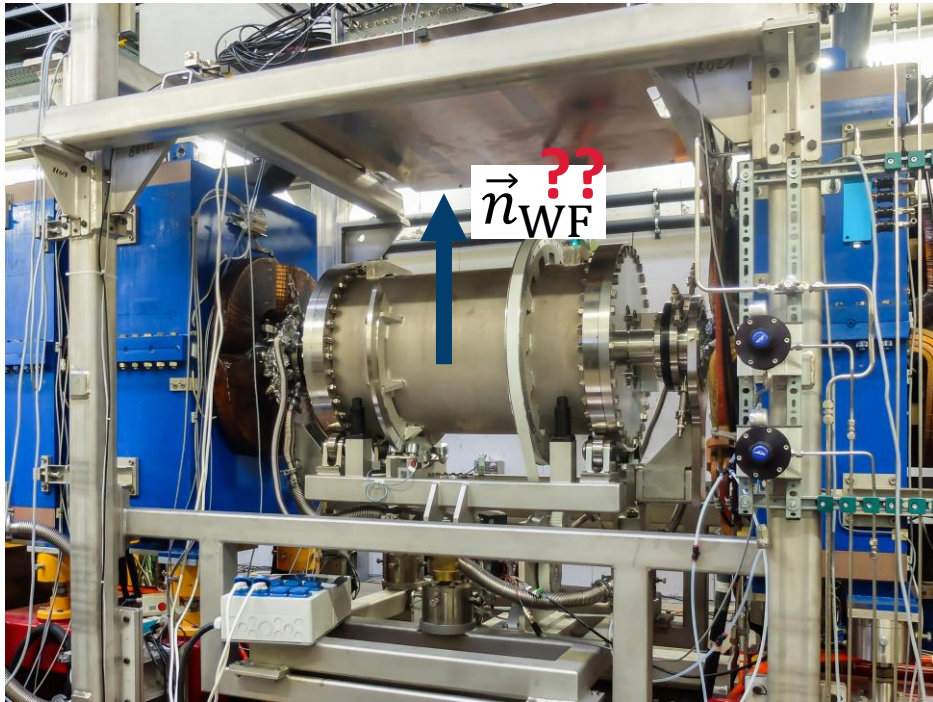


Blue Bunch



Red Bunch

A POSSIBLE SOLUTION



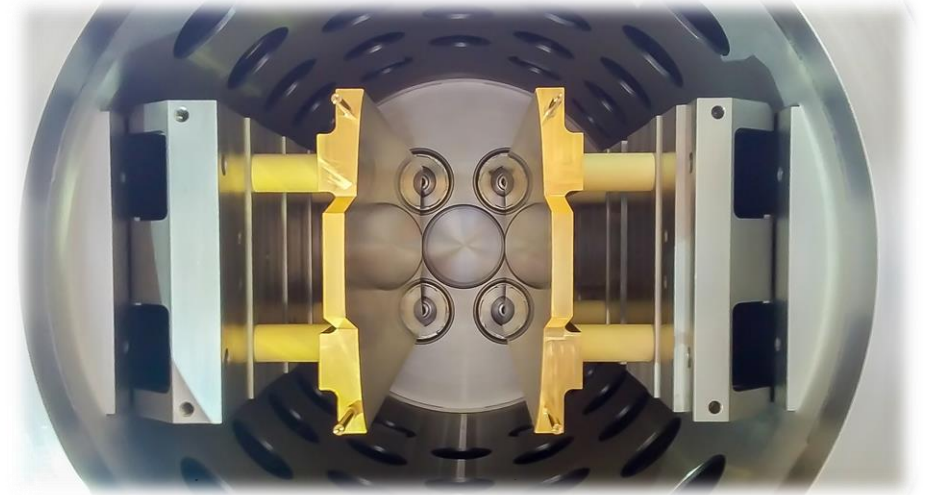
$$\begin{aligned}\epsilon^2(\phi^{\text{WF}}, \xi^{\text{Snake}}) &\propto |\vec{n}_{\text{WF}} \times \vec{n}|^2 \quad \vec{n}_{\text{WF}} : B \text{ field axis of rf Wien filter} \quad \vec{n} : \text{ISA} \\ &\approx \left| \begin{pmatrix} \phi^{\text{WF}} \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} \phi_0^{\text{EDM}} + \phi_0^{\text{COSY}} \\ 1 \\ \xi^{\text{Snake}} + \xi_0^{\text{COSY}} \end{pmatrix} \right|^2 \\ &\approx \left[((\phi_0^{\text{EDM}} + \phi_0^{\text{COSY}}) - \phi^{\text{WF}})^2 + (\xi^{\text{Snake}} + \xi_0^{\text{COSY}})^2 \right]\end{aligned}$$

- $\vec{B}$ - field direction of rf Wien filter is **only known from simulations**
- Depends on **geometrical design**

SIMULATIONS WITH FERRITES

- Initially, the RF Wien Filter was planned with a Ferrite Cage around the device
- Parallelism** of the **electrodes** has the largest influence on the **homogeneity** of the **electric field**
- Alignment of the **ferrite blocks** significantly impacts the **homogeneity** of the **magnetic field**
- Simulation of the **magnetic field** axis in which the effects of the **geometrical misalignments** (within a reasonable range) of the electrodes and the ferrite cage are studied are available

$$\text{Var} \left(\frac{\int_{-\frac{l}{2}}^{\frac{l}{2}} |\vec{H}_{\perp}| dl}{\int_{-\frac{l}{2}}^{\frac{l}{2}} |\vec{H}| dl} \right) \approx 1 \times 10^{-3}$$



Variable	Description	Distribution	Unit
x_1	Length of the upper electrode	$G(808.8, 0.1)$	mm
x_2	Length of the lower electrode	$G(808.8, 0.1)$	mm
x_3	Width of the upper electrode	$G(182, 0.1)$	mm
x_4	Width of the lower electrode	$G(182, 0.1)$	mm
x_5	Rotation of the upper electrode in the (xy -plane)	$G(0, 1)$	mrاد
x_6	Rotation of the lower electrode in the (xy -plane)	$G(0, 1)$	mrاد
x_7	Rotation of the upper electrode in the (xz -plane)	$G(0, 1)$	mrاد
x_8	Rotation of the lower electrode in the (xz -plane)	$G(0, 1)$	mrاد
x_9	Rotation of the ferrite structure in the(xy -plane)	$G(0, 1)$	mrاد
x_{10}	Rotation of the ferrite structure in the(xz -plane)	$G(0, 1)$	mrاد

EDM IN SM AND SUSY

